

Going against the Current:

The spatially heterogeneous impacts of the Programa Mais Educação

Presented and Defended by: David Figueroa *

Université Paris 1 – École d'Économie de la Sorbonne

Master 2 Development Economics

Supervised by: Clément Bosquet & Clément Gorin

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Abstract

Public policy evaluations routinely treat geography as a control variable, assuming that program availability equals effective access. This thesis challenges that assumption. I construct a multimodal road-river network of 14.9 million edges across Brazil to build a tier-based Index of Spatial Accessibility (ISA) from travel times, and use it to evaluate Brazil's Programa Mais Educação (PME), its largest extended school-day program. Interacting the ISA with treatment in a Propensity Score Matching Difference-in-Differences design reveals heterogeneity hidden in standard aggregates: while replicating PME's overall negative effect on mathematics, I find isolated schools suffer short-run declines more than twice as severe as highly connected ones. The mechanism is a staffing bottleneck, enrollment rises faster than teaching capacity, and the strain is worst where remote teacher labor markets are thinnest. Crucially, the apparent harm to connected schools largely reflects two biases, the differential attrition of weaker controls and the demographic decline of the areas treated schools occupy, rather than genuine program effects; once corrected, connected-tier effects converge toward zero while isolated effects persist, widening the spatial gap. The Amazon illustrates these extremes, but the lesson generalizes: aggregate evaluations that ignore physical accessibility risk severely distorting program assessments.

Keywords: Extended school day; Programa Mais Educação; Spatial accessibility; Multimodal transport networks; Amazon; Heterogeneous treatment effects; PSM-DiD;

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Contents

1	Introduction	6
2	Related literature	8
2.1	Extended School Days in Developing Countries	9
2.2	Brazil's Programa Mais Educação	9
2.3	Accessibility: From Euclidean Distance to Multimodal Networks	11
3	Data	12
3.1	School Data	12
3.2	Transportation Data	13
3.2.1	Rivers	13
3.2.2	Roads	14
3.2.3	Unifying the networks	15
3.3	Accessibility index	16
3.3.1	Time as accessibility	16
4	Empirical strategy	18
4.1	Base PSM-DiD	19
4.2	Heterogeneity by Accessibility	20
5	Results	22
5.1	Base specification results	22
5.2	PME & accessibility Results	24
5.3	Mechanism testing	26
5.4	Decomposing attrition	28
5.4.1	Mechanisms: Structural Protection and Area-Level Decline	28
6	Robustness Check	32
6.1	Alternative ISA indexes	32
6.2	Extended sample ISA-tier specification	34
6.3	Spillover and area-decline robustness	35
7	Conclusion	36
8	Appendix	45
A.1	Schools receiving PME funds per year	45
A.2	Fuzzy RDD design specification and results	45

A.3	HydroSheds river processing	48
A.4	Speed parameters and time penalties by edge type in the multimodal Amazon transportation network.	48
A.5	Road connector example	49
A.6	Brazil-Amazon graph network visualizaton	49
A.7	Spatial Imputation for Off-Network Schools	50
A.8	Basic Health Accessibility Index	51
A.9	Composite Weighted Accessibility Index	51
A.10	School log of Travel time to hospital distribution across ISA tiers	52
A.11	Pairwise Standardized Mean Differences (SMDs) Across Tiers: Full Sample	53
A.12	Pre-matching base specification mean testing: schools that ever participated in PME vs. schools that never participated	53
A.13	Fuzzy Differences-in-Discontinuity design specification and results	55
A.14	Variables used for the PSM-DiD specification	57
A.15	PSM Matched and Discarded observations: Base and Within-Tier Specifications	58
A.16	Propensity Score Balance Comparison: before and after.	59
A.17	Base specification Post-Matching Covariant Balance	59
A.18	ISA specification Post-Matching Covariant Balance	61
A.19	Base Specification Event Study	63
A.20	ISA-tier interacted specification Event Study	64
A.21	Heterogeneous PME effects by Accessibility tier (Extended sample version)	65
A.22	Heterogeneous PSM-DiD, coefficients as ISA tier differences	66
A.23	<i>Pooled Matching</i> Heterogeneous PME effects by Accessibility tier	68
A.24	Mechanism testing: Pooled ATTs on schooling inputs	69
A.25	Mechanism Analysis: Accessibility Tiers vs. Continuous Gradient	70
A.26	Differential attrition: Rates and regression tests	71
A.27	Attrition Tests: Spatial spillover robustness	71
A.28	Compositional imbalance: exposure to predicted school closures	72
A.29	Attrition testing: School network shocks and labor reallocation by accessibility tier	73
A.30	Attrition testing: Spatial Closure Effects by Tier on Schooling Inputs	74
A.31	ISA tier regressions with school closure interactions	76
A.32	ISA-tier interacted with extended sampe specification Event Study	78
A.33	Attrition Tests: Specifications and Estimation	79
A.33.1	Sample and notation	79

A.33.2 Mechanism 1 (Panel A): Did PME prevent structural closures?	79
A.33.3 Mechanism 2 (Panel B): Did open schools dodge the exam?	80
A.33.4 Mechanism 3: Spatial spillovers	80
A.33.5 Connecting the math to the intuition	82
A.33.6 Sample sizes across the mechanism pipeline	82
A.33.7 Standard error treatment	83

1 Introduction

A long-standing assumption in public policy evaluation is that program availability translates into program access. Governments transfer resources, schools receive them, and students benefit. In contexts defined by dense road networks and short distances to services, this assumption is largely harmless. In the Amazon basin, it is not. Over 25,000 kilometers of navigable waterways, a sparse and sometimes disconnected road network, and vast distances between population centers mean that two schools within the same municipality can face travel times that differ by orders of magnitude, one a twenty minute drive from the municipal seat, the other an eight hour boat journey. Standard program evaluations, conducted at the school or municipal level, treat these schools as equivalent. This thesis argues they are not, and that the failure to account for physical connectivity has systematically distorted our understanding of how public programs perform across diverse geographic landscapes.

To make this argument empirically, I construct a novel multimodal transportation network for the Amazon and the whole of Brazil that integrates road and river systems, expanding on the preliminary works of Rocha et al. (2023). Using Graph Theory, I model actual time-cost frictions by assigning differential speeds to road segments by surface type, incorporating river navigability data from HydroSHEDS, ANTAQ, and OpenStreetMap, and linking the two networks through formal ports, road-river crossings, and informal riverbank landings. Unlike approaches relying on Euclidean distance or road-only proximity (Fortney et al., 2005; Guerrero et al., 2020; McGuire et al., 2021), this network embeds the physical realities of Amazonian geography to compute each school’s effective connectivity to different Points of Interest (POI). From this network, I derive an Index of Spatial Accessibility (ISA) that partitions schools into low, middle, and high connectivity tiers.

The natural laboratory for testing whether this index captures something real is Brazil’s *Programa Mais Educação* (PME). Launched in 2007 and expanded significantly over the following decade, the PME extended the school day to a minimum of seven hours through supplementary activities in mathematics, Portuguese, sports, and the arts, reaching over 60,000 schools by 2014 (MEC, 2010). Despite its scale and its role as a central pillar of the National Education Plan, the program’s evaluation record is one of near-uniform null or negative effects on academic outcomes (Almeida et al., 2016; Vidigal & Vidigal, 2018; Xerxenevsky, 2012), a pattern the literature has documented with reasonable consistency but without satisfactorily explaining its distribution across space. The PME is thus not simply a program of interest, it is a program whose discouraging aggregate results demand a spatial unpacking.

That unpacking emerges once connectivity is incorporated. Replicating a Propensity Score Matching with Difference-in-Differences (PSM-DiD) design, then interacting it with ISA tier, I show that PME's effects are not uniform but systematically structured by physical accessibility. In the headline results, schools in the lowest accessibility tier experience short-run declines in mathematics more than twice as severe as those in the highest tier. The proximate mechanism is a friction of implementation: PME raises enrollment faster than schools can expand their teaching staff, straining instructional capacity. This pressure is present across the accessibility distribution, but its consequences fall hardest where remote teacher labor markets are thinnest. In the least accessible schools, a given staffing shortfall cannot be met by drawing on a deep local pool of qualified teachers, and the same travel frictions that define isolation plausibly raise teacher absenteeism. Neither channel is fully traceable in the staffing variables themselves, but both would compound the initial implementation shock and are consistent with the persistence of negative effects in low-accessibility schools beyond the point at which staffing catches up. The program is, in effect, the same intervention applied to two very different logistical realities, and the transportation network largely determines which reality a school inhabits.

A second and more subtle finding concerns the apparent negative effect among well-connected schools, which closer analysis shows to be substantially an artifact of two biases that the standard design cannot detect. First, PME-treated schools are significantly less likely than their matched controls to disappear from the panel, so the surviving comparison group is positively selected and the estimated effects are biased downward. Second, treated schools are disproportionately located in areas undergoing demographic decline, for which nearby school closures serve as a proxy following Lehtonen (2021). And it is this decline, rather than the program, that drives much of the connected-tier result. Once both biases are accounted for, the negative effect in high-accessibility schools converges toward null, particularly over the medium term, while the effect in isolated schools persists and, if anything, strengthens. The accessibility gap is therefore not an artifact of where schools happen to be located: correcting for these biases widens it rather than narrowing it.

These results carry implications beyond the PME. Aggregate national or state-level treatment effect estimates are likely incomplete when geography is ignored: they simultaneously underestimate a program's potential in connected hubs and overstate its viability in disconnected hinterlands. Without complementary investments in transport and teacher supply, financial transfers face a *last mile* problem that no evaluation design can resolve by controlling for it.

This thesis contributes to three bodies of literature. First, it advances the literature on transport geography and public service provision in the Amazon (Parry et al., 2018; Pera

& Datta, 2018), demonstrating that the “friction of distance” in a multimodal network is not merely physical but a proxy for state capacity, and that this friction can be credibly measured and used causally. Second, it contributes to the heterogeneous treatment effects literature (Athey & Imbens, 2017; Merfeld, 2019) by explicitly modeling the transport network as the transmission mechanism for policy effectiveness, rather than absorbing geography into fixed effects. Third, it provides spatially nuanced evidence on extended school day programs in developing countries (Almeida et al., 2016; Holland & Li, 2015), showing that effectiveness is not automatic but conditional on the logistical environment into which a program is introduced. Across these contributions, the central claim is the same: the infrastructure gap in the Amazon is also an educational gap, and treating the two as separate problems is part of why evaluations keep finding null results.

The remainder of this thesis is organized to trace this spatial argument from theory to empirics. Section 2 situates the PME within the broader literatures of extended school days and transport geography, while Section 3 details the construction of the multimodal network. The empirical framework and the PSM-DiD specifications are developed in Section 4, followed in Section 5 by the primary results, the teacher-supply mechanism, and the differential-attrition and area-decline analysis that reframes the connected-tier effects. Section 6 then confirms that these patterns are not artifacts of buffer choice or index construction. Finally, Section 7 concludes by examining the implications of these results, stating the case for designing geography-conscious public programs.

2 Related literature

This thesis requires three bodies of literature to work simultaneously, and each of them, on its own, is incomplete in a way the others can address. The extended school day literature has established that implementation environment matters but has rarely been able to say what that means in concrete geographic terms. The PME-specific literature has documented the program’s near-uniform failure with reasonable consistency but has treated the spatial dimension as something to control for rather than something to explain. The accessibility literature, meanwhile, has developed tools to measure geographic connectivity with precision, but those tools have been applied almost entirely to health access and poverty mapping, not to education evaluation. The argument of this thesis lives at the intersection of all three. What follows traces what each contributes, and where the gaps lie.

2.1 Extended School Days in Developing Countries

The policy logic behind extended school day programs is straightforward: if students spend only four hours a day in school, adding structured time should expand opportunities for learning and reinforcement. Several countries in Latin America acted on this logic from the early 2000s onwards: Chile's *Jornada Escolar Completa* (Bellei, 2009), Colombia's extended day (Hincapié, 2016), Uruguay's full-time school program (Cerdan-Infantes & Vermeersch, 2007), and Peru's *Jornada Escolar Completa* (Agüero et al., 2021). The common thread across these evaluations is that effects are modest, heterogeneous, and conditional on something beyond the hours themselves.

Holland and Li (2015), reviewing 19 studies across Latin America and the Caribbean, find that while there is some tendency toward positive impacts on test scores and secondary outcomes such as crime reduction and teenage pregnancy, the effects are small and uneven. Bellei (2009) and Hincapié (2016) find little effect in Chile and Colombia respectively. Agüero et al. (2021) find larger gains in Peru, but in a program that combined extended hours with significant curricular restructuring, it goes beyond simply adding time. The World Bank (2024) review reaches a similar conclusion: the relationship between additional hours and improved test scores is generally weak, with clearer gains appearing in programs that transform the curriculum rather than merely lengthen the day. Cost-effectiveness analyses further suggest that extending the school day is among the most expensive interventions per unit of learning gain.

What these evaluations collectively point toward is that the quantity of added time matters less than the environment into which that time is introduced. A program that functions smoothly in a well-resourced school near an urban center may impose organizational costs that overwhelm any pedagogical benefit in a school that lacks the staff, infrastructure, or logistical support to absorb the expansion. Brazil's own program history offers a useful illustration of this point. Blulm and Giuberti (2024) evaluate the successor program *Novo Mais Educação*, which concentrated additional hours specifically on mathematics rather than the broader range of activities that characterized the original PME, and find more positive results. That a leaner, more focused program design produced better outcomes is, at minimum, consistent with the idea that implementation complexity was part of what made the original fail, a pattern the Brazilian case will make more concrete.

2.2 Brazil's Programa Mais Educação

Brazil's *Programa Mais Educação* (PME), established in 2007, represents one of the largest extended school day programs implemented in the developing world. The program ex-

tended the school day to a minimum of seven hours through supplementary activities including pedagogical support, sports, arts, digital culture, and health promotion in the counter-shift (MEC, 2010). By 2014 it reached over 60,000 schools and was a central pillar of the National Education Plan.

The roll-out was phased and non-random. Initially, the program prioritized schools in metropolitan areas and low IDEB scores. Over time, the proportion of Bolsa Família (BF) recipients among enrolled students became an increasingly important selection criterion, eventually formalizing in 2012 into a 50 percent threshold that created a discontinuity exploitable for causal inference.

The evaluation record is, with near unanimity, discouraging. Almeida et al. (2016), in the most prominent assessment conducted with the World Bank, use PSM-DiD on the 2007–2011 rollout and find no significant impact on dropout rates and negative average effects on math scores. They attribute this finding to the challenges of initial nationwide implementation, linking it to the disruption imposed on the established habits of students, teachers, and schools, which they then support with evidence that schools joining the program in 2010 showed smaller negative effects than 2008 cohorts, consistent with gradual consolidation. This pattern of negative results on scores recurs across the literature: Xerxenevsky (2012) finds similar conclusions for Rio Grande do Sul. Aquino and Menezes-Filho (2015) report math performance reduced by approximately 7 points on the Prova Brasil scale in the Midwest. Vidigal and Vidigal (2018) find null effects over a broader time window. The Fundação Itaú Social (2015) corroborates all of the above, finding an initial negative math effect that dissipated over time with no significant impact on Portuguese or dropout. De Oliveira and Terra (2016) and de Resende et al. (2020), using fuzzy regression discontinuity designs (FRDD) on the later program years, find more credibly identified but still modest effects. The consistency of these results across methodologies, time periods, and regions is notable, showing this is not a case of one or two anomalous studies, but a literature that has repeatedly arrived at the same finding from different angles.

What is worth highlighting is that none of these studies seriously engages with the spatial dimension. Geography appears in the existing literature as a fixed effect, a regional control, or simply a description of where the data comes from. The implicit assumption is that formal participation in PME adequately characterizes treatment, that a school receiving PME funds is, in any meaningful sense, a PME school. In a region where two schools within the same municipality can face travel times that differ by hours, one a twenty minute drive from the municipal seat, the other an eight hour boat journey, that assumption is difficult to sustain. The question the literature has not asked is whether

the implementation friction that Almeida et al. (2016) identify as the mechanism is itself distributed unevenly across geography, and if so, whether the aggregate negative effects are obscuring something more heterogeneous underneath.

2.3 Accessibility: From Euclidean Distance to Multimodal Networks

To ask whether implementation friction is geographically structured requires a way of measuring connectivity that goes beyond administrative boundaries or simple distance. In the Amazon basin, this is not a minor methodological refinement, it is a precondition for the analysis. With over 25,000 kilometers of navigable waterways, a sparse and sometimes disconnected road network, and vast distances between population centers, the region defies the assumptions embedded in standard accessibility measures (Becker, 2013). For riverside communities, rivers may be the primary (and sometimes the only) means of reaching schools, health facilities, and markets. Pera and Datta (2018) document how transport costs in the Amazon can be orders of magnitude higher than equivalent distances in Brazil's Southeast. Parry et al. (2018) show that river-dependent communities face systematically worse access to education, health, and government services. Santos de Lima et al. (2024) trace two decades of drought impacts, showing how extreme low-water periods routinely isolate populations in ways that are entirely invisible at the municipal level.

Standard accessibility measures cannot capture this. Euclidean distance is a poor proxy for actual travel costs in many settings: Fortney et al. (2005) demonstrate that it systematically underestimates travel burden and misidentifies underserved populations. McGuire et al. (2021) show that replacing it with network-based travel times substantially alters program evaluation conclusions, with Guerrero et al. (2020) finding similar results for conditional cash transfers in developing countries. In the Amazon context, a road-only network is nearly as misleading as Euclidean distance, since it simply ignores the river system that a large share of the population depends on for mobility.

The concept of accessibility has been a foundational concern in transport geography since Hansen (1959). Earlier applied work attributed travel frictions through two-step floating catchment areas (Luo & Wang, 2003), with subsequent extensions incorporating multi-modal transport networks (Lin et al., 2018). More recently, graph theory has made it possible to model transportation networks as weighted graphs, with shortest-path algorithms yielding travel time matrices that capture the actual friction of the network rather than the fiction of straight-line distance (El-Geneidy et al., 2016; Sarkar et al., 2021). These tools have been developed and refined primarily in health geography, where the stakes of

access measurement are high and the data demands are manageable.

The one study that has applied this approach to the Amazon is Rocha et al. (2023), who construct the first multimodal network for the region combining OpenStreetMap roads with river routes derived from HydroSHEDS and GloRiC, demonstrating that catchment areas around health facilities expand dramatically when rivers are incorporated. Their contribution is significant, but limited in two respects: it covers only the state of Amazonas, and it is descriptive, as the network is used to characterize access patterns, and not yet to estimate any causal effects.

The gap this thesis addresses sits at the intersection of these three literatures. The extended school day literature has the outcome variables and the policy question. The PME literature has documented the failure but lacks the spatial tools to explain it. The accessibility literature has those tools but has not applied them to education evaluation or to causal identification. No existing study combines a multimodal Amazon transportation network with a credible identification strategy for estimating spatially heterogeneous treatment effects of an education program. That combination is what this paper attempts.

3 Data

3.1 School Data

The data relating to school can be divided into three sections: school characteristics data, SAEB results data, and program participation data. Together, these three sources form the foundation for the base specification, developed in the Empirical Strategy section, which abstracts from geographic consideration. The period of study will comprise 2005 until 2015 (specific year-intervals depending on the empirical strategy), which are the years that allow to check the pretrends, and extend to before the *Programa Novo Mais Educacao* was implemented in 2017, which differs in objectives. Although data from before would be desirable in order to further check for pre-trends, a methodological change happened during 2005, where the SAEB became a census-wide test, and only after standardized data was logged to schools explicitly (BRASIL. Ministério da Educação, 2005). Data from before would be best for the parallel trends assumption, and so this is one limitation for this study which will be described more in detail in the Empirical strategy section.

The school characteristics data were collected from the *Censo Escolar* (INEP, 2026). The data was collected yearly starting from 2007 up until 2015, and then joined keeping only characteristics such as students age, quantity, race, teachers, schools infrastructure, among other things. This database held information on approximately 150,000 schools over this

period. What is now needed are test results, the best option are the *Prova Brasil* tests.

The *Prova Brasil* results were collected from the INEP (2025). The results are biyearly and at the school level, taken from 2007 until 2015. The data includes data on math, Portuguese, and average notes, as well school IDEB scores. The data includes results for 40,000 schools on average per year. However, the data on these schools is unbalanced, since some schools cease to exist along the way.¹ To put it in perspective, only around 19,000 schools remain from the period from 2007 to 2015. The final piece for the non-geographic specification would be program participation and selection criteria.

The data on PME participation and PBF was solicited directly from the Brazilian government and is in different databases. The PME database comprises yearly school-level transfers from the *Programa Dinheiro Direto na Escola* to the schools that were PME participants that year. Schools that participated in PME per year can be seen in Appendix A.1. The PBF data, which comprise students that are part of the PBF were then matched to the *censo escolar*'s number of students per school, to get the school's percentage of PBF students. However, as mentioned in the Appendix A.2 percentages needed to be treated as some regularities presented themselves.

3.2 Transportation Data

In order to incorporate geographic exogeneity into the specifications I first needed to elaborate a transportation network for Brazil and the Amazon. To create this network there were two different sources types: Roads and Rivers. Each of them required different treatment both from the source data, to the transformation into the full graphed network.²

3.2.1 Rivers

Rivers and raw water bodies data came from HydroSHEDS, with two relevant datasets that contained the same water bodies vector layers: HydroRivers (Lehner & Grill, 2013) and GloRiC (Ouellet Dallaire et al., 2019). Both datasets are useful when jointly used due to the connected, next down nature of the data, where it is possible to track the sequence of rivers. Moreover, the classification of rivers, as well as the data on river variability and discharge was essential to preserve the relevant rivers. This being said, by themselves, the GloRiC and HydroSHEDS database do not necessarily differentiate water bodies that are navigable from those that are not, such as underground water bodies. Therefore, more

¹I tested the effect of the program on attrition. I enter into the details in Section 5.4

²All the data relating to transportation was mostly treated in R using the libraries: sfnetworks, igraph, tidygraph, lwgeom, and dplyr.

steps needed to be taken in order to make a fully navigable dataset of the Amazon.

Starting from the rationale of Rocha et al. (2023), filtering *possibly navigable* river routes by excluding low discharge and/or very high variability (as classified by HydroSHEDS) is a plausible starting point. However, in addition to the classification of water bodies of HydroSHEDS, I used two additional sources to complement the navigable elements of the HydroSHEDS layer: ANTAQ and OSM. First, the ANTAQ (2022) vector layer contains confirmed navigable routes by the National Agency of waterbody transport (ANTAQ for its name in portuguese), which are kept by routing the HydroSHEDS layer with the overlay of the ANTAQ database. These routes will be kept independent of their *Variability/Discharge* category in the HydroRivers database due to their fact checked nature (see Appendix A.3 for a visual representation). Then, by scrapping the OSM rivers from the Geofabrik (2024) database, an additional, but less robust vector layer is extracted. Within the OSM scrapping, some attributes contained tags such as "Navigable: No" or "Navigable: Yes". Those attributes that contained this categories were consequently dropped or kept respectively regardless of the HydroSHEDS category. Once this filtering was done, the routing process was done as with the ANTAQ database. Nevertheless, the routed OSM layer was susceptible to the *Variability/Discharge* filtering. Finally the river speeds joined to the network are those of Moura (2017) which are also those used by Rocha et al. (2023).

A navigable river network, however, only gets you as far as the water's edge. Connecting river routes to the schools and service points that are the actual origins and destinations of interest requires a road layer.

3.2.2 Roads

For the base Roads layer, the data was once again fully extracted from the Geofabrik (2024) database. The categories of roads kept were motorways, trunks, primary, secondary, tertiary, residential streets, as well as some pedestrian and cycling roads. From this scrap, roads with categories such as "abandoned" "proposed" and "construction" were discarded. An additional procedure was done to the base OSM extraction. Given the need for the road system to be fully connected, an additional procedure was applied³: an algorithm created gap-filling edges of up to 2 kilometers wherever roads fell short of connecting, assigning these the slowest road categories ("unclassified" and "unpaved") as a conservative default (Visual example is in Appendix A.5). Each road in the full network was then assigned a travel speed regulated by surface type, with paved and unpaved roads receiving

³In the Amazon mostly, some roads (edges) required minor roads that were not included, or were drawn slightly shorter than needed in order to connect the roads

a bonus and penalty respectively (full speed assignments and penalties by road category are detailed in Appendix A.4). Finally, schools and points of interest (POI) such as hospitals, food businesses, and administrative buildings were snapped to the road network via a connector algorithm, with the resulting edges assigned walking speeds of 5 km/h. With these components in place, the road network was complete.

3.2.3 Unifying the networks

Due to the nature of the HydroSHEDS database, the bodies of water tend to be independent of the road infrastructure that might or might not be present. In many cases, it is the same for the land infrastructure, where connection to rivers might not be accounted for (e.g. road-to-water not necessarily happening at a ferry port). Consequently, in order to connect both, three distinct types of connections were created, each with its own sets of "transfer costs": Port landings, Road/river crossings, and informal landings. Port landings was the most straightforward type. First, official ports and ferry landings were extracted from the Geofabrik (2024) database. Then an algorithm created maximum 5 km edges from the rivers to these ports. The changing of land to river generally incurs additional time costs to transfer from one to the other, depending on the formality/informality of the landing. Thus, *transfer time costs* were assigned to this and the other two connectors, which are visible also in Appendix A.4⁴. Next was the road/river crossings. The procedure was to create a *transfer point* wherever roads and river edges crossed, under the assumption that it is likely to be able to transfer to land or viceversa wherever a bridge might be (accounting, with its possible transfer times). Finally, an additional edge creating algorithm was run to create informal landings. Since river transportation in the Amazon can be done in big part not through formal ports but through landing in river banks, it is essential to account for possible routes where boats might land to transfer to roads (ANTAQ, 2018). The criteria of the algorithm was to find the nearest road from the river every 500 meters. Additionally, this type of connector had the slowest of transfers speeds and an additional transfer cost which could account for landing the boat, and changing into land.

One additional type of POI connector was elaborated specifically for schools to rivers, which was made to account for the most isolated schools in the Amazon region. Since the previous POI connectors and edges created rely on roads as an intermediary, schools that do not have roads in the vicinity or which have scarcely have had edges created in

⁴This threshold was large due to the width some Amazon rivers may attain (up to 50 kilometers). Nevertheless, a possible improvement would be to account for river banks instead of single river linestrings in order to be more precise

OSM for them would be, keeping only the previous logic, importantly unaccounted for. With this in mind, the edge creating algorithm - that was only present in the Amazon region - had a criteria of making a maximum of 50km edge to the river. This criteria, more than a literature-backed criteria, was an empirical fix due to the lacking presence of consistent routing for certain isolated schools in the region. Therefore a further refinement of this database would be the substitution of these connectors once more edges are created around the most isolated schools in the Amazon. With this techniques combined, 39,454 out of 40,137 schools that had Prova Brasil results and were geolocated had been embedded into the network. The remaining 683 schools were left disconnected as expanding parameters to include them in the network would have significantly raised the probability of creating artificial edges in the network that did not reflect the dynamics of the real transportation network. The specific treatment of this schools will be elaborated in the next section.

Now, with the all edge components set, the last procedure was the splitting of the edges where the nodes of the new POI lines were drawn such that the edges would be fully connected. In total, this network accounts for 1.31 million kilometers of roads, and over 300 thousand kilometers of rivers in the Amazon, without accounting for cyclable, walkable, minor, and smaller unclassified paths, which in it of themselves were shy of 2.8 million kilometers. To connect both the road and river network, this required around 5 thousand kilometers of connectors between roads and Points of Interests (Hospitals, food centers, and administrative centers), 7.6 thousand kilometers of connections between roads and rivers, and around 7 thousand kilometers of path account for informal landings between roads and rivers. In graph terms, the final network was comprised of 14.88 million edges and 10.93 million nodes (a visual representation is in Appendix A.6). Although computationally expensive, with this network we can now compute travel-time based accessibility measures for each school in the sample.

3.3 Accessibility index

In order to measure the 'accessibility' of the schools, one key question that first has to be asked is: *the schools accessibility in relation to what?*. For the purpose of this study, I will focus on one aspect of accessibility: effective time-cost.

3.3.1 Time as accessibility

The indexes relating to time-cost are based on the idea of *location-based* accessibility, notably definted by Geurs and Van Wee (2004). Here, they define accessibility more by the

time it takes from individuals in schools to access other POIs, which in this context are as hospitals, food centers, and administrative centers. I take this definition as it is the one that balances data accessible in the Amazon context, as well as encapsulates, through the presence of other *locations* how a school relates to other desirable activities to do. This definition is also relevant because of the potential mechanisms regarding teacher supply and teachers willingness to relocate that will be further elaborated on the results section.

In a more practical note, although the variable of time could be left to be continuous, I will elaborate the index based on 3 levels: low, middle and high. This decision is made since both indexes relating to time include hospitals (one having hospitals as only POI and the other weighting with other POI's), since they are fully mapped by the Brazilian government. Quite notably, the proximity to a hospital can have life-changing implications. Therefore, the groups of accessibility will be defined on measures of what the distance to the hospital imply in terms of medical consequences. More specifically, I use the literature of what the lower-bound standard should be (Bosanac et al., 1976), thus defining the *High* accessibility, and what the upper-bound beyond which serious medical complications could arise (Alkire et al., 2015). From this I create the *Low* accessibility group. More details of this can be found in Appendix A.8 & A.9. The primary accessibility analysis is restricted to a strict sample, explicitly excluding the 683 schools without formal network connections. This sample restriction serves a dual purpose. Methodologically, it avoids relying on the ex-post spatial imputations used to artificially connect the network (detailed in Appendix A.7). Econometrically, it guarantees a conservative approach, because including the imputed schools generates upper-bound regression results (further discussed in the Data section), utilizing the strict sample safely isolates the lower-bound estimates. Hereinafter, all analyses refer to this strict sample unless otherwise specified.

In Appendix A.10, we can see the distribution of schools by tier. It can be seen that the high tier accumulates most of the school's around the lower end of their threshold, whereas the medium tier is relatively balanced in the distribution of its observations. Lastly, the Low tier accessibility also seems to concentrate most of its observations on the 120-240 minute timeframe, but continues to have schools whose distance is up to 12 hours and more from the nearest *Unidade básica de saúde* (UBS).⁵ Moreover, the differences in characteristics between the ISA tiers can be found in Appendix A.11. What is important to note in this table is that Low and Medium accessibility schools are similar on many variables except some such as IDHM, on mathematic scores and having science laboratories, whose SMD value differences are still quite small by conventional standards. So

⁵A UBS serves as the primary care gateway for Brazil's public health system. It provides routine medical consultations, preventative care, and basic treatments, but is not equipped for emergency services or complex surgical procedures.

between these categories, it could be argued that the main difference would be their accessibility. Nevertheless, the differences between High tier compared to both Low and Medium tiers respectively do present significant differences in various categories, showing the systematic differences that might be at play for more accessible schools compared to their counterparts.

That these differences exist across tiers confirms the index is capturing real variation in school environments rather than noise. Whether that variation translates into heterogeneous program effects, however, is an empirical question that requires a careful identification strategy given that PME itself was not randomly assigned.

4 Empirical strategy

We seek to analyze the effect of the PME on the test performance of the ensino fundamental students. However, to simply compare treated versus untreated schools would lead to heavily biased results, since as shown in Appendix A.12, these two groups are compositionally different on almost all aspects that we could control for. This has to do with the fact that the PME targeted specifically schools that had lower IDEB scores (a measure composed of average SAEB/Prova Brasil scores and student promotion rates) and regions that were marked with social and educational vulnerability (MEC, 2010).

Various empirical strategies could circumvent the differences in schools that could get into the program, each unveiling slightly different insights. For the purposes of this thesis, I will go in-depth into the PSM-DiD, as it focuses on the impacts of PME on scores since the beginning of the implementation.⁶ To answer the question on the effect of accessibility on the programs effects, the PSM-DiD specification will be done in two parts. First, there will be a “base” reproducibility-aimed specification looking to replicate results of previous literature as a form of robustness test. The second part will more directly test the hypothesis of the paper by then interacting the “base” specification with the constructed measures of accessibility. It is worth noting that the PSM-DiD reproduction exercise will use years 2005 to 2011 only. Starting 2012, clear cut thresholds where schools could now apply were announced, among these thresholds there was one relating to the schools Programa Bolsa Familia (PBF) recipients being over 50% of school students, or if the school had under 4.2 or 3.8 IDEB scores for elementary and middle school respectively, changes

⁶Two additional identification strategies were tested but are not the focus of this thesis: A Fuzzy Regression Discontinuity (FRDD) as replication exercise (details in Appendix A.2). Moreover, a Fuzzy Differences-in-Discontinuities (FDiffDisc) was also tested (details in appendix A.13). Both yielded non-significant results

that would make the comparison of 2007 to 2013 and onwards more susceptible to affecting unobservables. With this being said, I now describe the two PSM-DID specifications.

4.1 *Base PSM-DiD*

The base specification of the PSM-DiD in the context of PME has been previously used by Almeida et al. (2016) and Xerxenevsky (2012). The key idea of this method is, before running a DiD, where the parallel trends assumption between control and treatment would not hold, the PSM matches treated and untreated schools on the basis of observables. Rather than matching variable by variable, it matches on a propensity score, defined as the conditional probability of treatment given a vector of pre-treatment covariates $\mathbf{X}_s$ (measured in 2007 for most variables and 2010 for the IDHM):

$$p(\mathbf{X}_s) = \Pr(D_s = 1 \mid \mathbf{X}_s). \quad (1)$$

The score is estimated by a Probit specification:

$$\Pr(D_s = 1 \mid \mathbf{X}_s) = \Phi(\mathbf{X}_s' \boldsymbol{\beta}), \quad (2)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The covariate vector $\mathbf{X}_s$ includes (i) pre-trend variables computed as the 2005–2007 change in SAEB Mathematics, SAEB Portuguese, and IDEB scores; alongside a comprehensive set of baseline characteristics measured in 2007, comprising: (ii) levels of the aforementioned educational outcomes; (iii) school-size and staffing variables (enrolment in fundamental education, number of classrooms used and existing, total staff); (iv) infrastructure indicators (sports facilities, science and computer laboratories, teachers' room, multimedia equipment, sewage connection, school-meal provision); (v) administrative type (state-managed dummy); and (vi) municipal characteristics (log population, IDHM). The choice of covariates follows closely the specification of Almeida et al. (2016). The full variable list and descriptions can be found in Appendix A.14. The control and treatment groups differ slightly from those of Almeida et al. (2016) in that schools that closed during the study period are retained, on the grounds that their exclusion would systematically omit underperforming schools and bias the estimates.

Schools were matched with replacement, prioritizing match quality over independence of observations. To enforce the common support assumption, treated and control observations whose propensity scores fall outside the overlap region are discarded; the resulting sample sizes and number of discarded observations are reported in Ap-

pendix A.15. The overlap before and after matching is shown in Appendix A.16, and post-matching covariate balance is confirmed in Appendix A.17.

Once treated schools are matched, the ATT is estimated from the following DiD specification:

$$Y_{s,t} = \alpha_s + \gamma Post_t + \tau (PME_s \cdot Post_t) + \varepsilon_{s,t}, \quad (3)$$

where α_s is a school fixed effect absorbing all time-invariant school characteristics, PME_s indicates whether the school was a PME beneficiary, $Post_t$ is a post-treatment indicator equal to one for years $t \geq t^*$, and τ is the coefficient of interest. The regression is estimated by weighted least squares using the matching weights ω_s . The estimator of interest is:

$$\hat{\tau}(t^*) = \left(\bar{Y}_{1,t^*} - \bar{Y}_{1,2007} \right) - \left(\bar{Y}_{0,t^*} - \bar{Y}_{0,2007} \right), \quad (4)$$

where $\bar{Y}_{d,t}$ denotes the matching-weighted mean of $Y_{s,t}$ in group d . Estimating this separately for each post-treatment horizon $t^* \in \{2009, 2011\}$ and each outcome $Y \in \{\text{SAEB Math, SAEB Portuguese, IDEB}\}$ yields a 3×2 matrix of ATT estimates, constituting an event-study of the program's dynamic effects constrained by the biennial frequency of SAEB measurement.

The causal interpretation of $\hat{\tau}$ rests on the Parallel Trends Assumption holding conditional on the matched sample. Given the biennial frequency of SAEB scores, only one pre-treatment period is available for validation: the 2005-2007 window, prior to any PME participation. I estimate a placebo version of Equation 3 using 2005 as the pre-period and 2007 as the post-period, a window during which no school had yet entered the program. As shown in Table 1, placebo ATTs are statistically indistinguishable from zero at the 5% level for both SAEB Mathematics and Portuguese across all specifications, supporting the plausibility of parallel trends. IDEB placebo results are included for completeness but interpreted descriptively, consistent with its treatment throughout this thesis. The full dynamic treatment path across the 2009 and 2011 horizons is presented alongside the results in Section 5.

4.2 Heterogeneity by Accessibility

The central hypothesis of this paper is that the effect of PME on student performance is attenuated in schools with poor multimodal accessibility, where the logistical friction of distance (which is most pronounced in the Amazon Basin) imposes a binding constraint

Table 1: Placebo test: pre-treatment trends (2005 vs. 2007)

	Base (Pooled)	ISA Tiers
<i>Panel A: SAEB Mathematics</i>		
Pooled Placebo ATT	0.407 (0.367)	- -
Low Tier Placebo ATT	-	0.508 (1.130)
Mid Tier Placebo ATT	-	-1.210 (0.972)
High Tier Placebo ATT	-	0.925* (0.540)
<i>Panel B: SAEB Portuguese</i>		
Pooled Placebo ATT	0.296 (0.397)	- -
Low Tier Placebo ATT	-	1.020 (1.120)
Mid Tier Placebo ATT	-	-0.647 (0.930)
High Tier Placebo ATT	-	1.050* (0.536)

Notes: The **Base** column reports the pooled DiD coefficient. The **ISA Tiers** column reports the absolute Placebo ATTs for each specific accessibility tier, calculated via linear combinations of the baseline and interaction coefficients, ensuring that treated schools are compared strictly against control schools within their respective tiers. Standard errors (in parentheses) are clustered at the school level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

on program implementation. To test this hypothesis, each school s is assigned to one of three terciles of the ISA constructed in Section 3, denoting $A_s \in \{\text{Low}, \text{Mid}, \text{High}\}$, and Equation (3) is augmented with treatment-tier interactions:

$$\begin{aligned}
 Y_{s,t} = & \alpha_s + \gamma Post_t + \sum_{k \in \{\text{Mid}, \text{High}\}} \eta_k (Post_t \cdot \mathbb{1}[A_s = k]) \\
 & + \tau^{\text{Low}} (PME_s \cdot Post_t) + \sum_{k \in \{\text{Mid}, \text{High}\}} \delta_k (PME_s \cdot Post_t \cdot \mathbb{1}[A_s = k]) + \varepsilon_{s,t},
 \end{aligned} \tag{5}$$

where the omitted accessibility category is Low. The η_k terms allow the post-period time shock to differ across accessibility terciles in the never-treated group, relaxing the assumption that control schools in Mid and High accessibility areas would have evolved on the same trend as Low accessibility controls absent treatment. The coefficient τ^{Low} identifies the ATT for treated schools in the lowest accessibility tercile, while δ_{Mid} and δ_{High} measure the *differential* treatment effect for schools in the Mid and High terciles relative to that benchmark. The tercile-specific ATTs are recovered as $\hat{\tau}^{\text{Mid}}(t^*) = \hat{\tau}^{\text{Low}}(t^*) + \hat{\delta}_{\text{Mid}}(t^*)$ and $\hat{\tau}^{\text{High}}(t^*) = \hat{\tau}^{\text{Low}}(t^*) + \hat{\delta}_{\text{High}}(t^*)$, with standard errors computed from the joint covariance matrix of the estimated coefficients. As before, Equation (5) is estimated by weighted least

squares using the matching weights ω_s , with standard errors clustered at the school level, and is run separately for each horizon $t^* \in \{2009, 2011\}$ and each outcome $Y \in \{\text{SAEB Math, SAEB Portuguese, IDEB}\}$.

Matching is performed separately within each accessibility tier, with post-matching balance reported in Appendix A.18. Few variables exceed the $SMD > 0.1$ threshold suggested by Rubin (2001); among those that do, two out of three belong to the Low tier, with a maximum SMD of 0.133. This reflects that in more isolated areas, PME tended to target schools with more classrooms in use relative to total capacity, possibly indicating overcrowding, and fewer multimedia resources, consistent with tighter budget constraints in these schools. As with the base specification, the parallel trends assumption is validated via the 2005–2007 placebo test reported in Table 1, with tier-specific placebo ATTs indistinguishable from zero at the 5% level for both main outcomes.

5 Results

5.1 Base specification results

First, the results of the *base* specification are shown in Table 2 (Visual representation can be seen in Appendix A.19). The short term Math results (2007 to 2009) go in a similar direction than those found by Almeida et al. (2016), being significant and negative, albeit being less negative than those found by them. This could be explained by those schools that were left out by not being in the full period. Those schools could have been closed down due to reasons other than their performance. Moreover, another interesting results is that we do find significant and negative results for Portuguese scores, which show to be of a similar but smaller negative effect than the Mathematics scores. When we extend the analysis to the medium term (2007 to 2011) we see a similar but decreasing pattern: Math results are negative and significant but to a lower extent, Portuguese is still negative but non-significant. Once again, these results follow Almeida et al. (2016)’s pattern.

With regards to IDEB scores, these results are shown more descriptively, as IDEB scores were part of the selection criteria, but they also work as an average of the scoring and approval rates. Therefore, the fact that it behaves similarly to the coefficients of the Math and Portuguese score adds to the results. What we now seek to observe is, if beneath these estimators there is heterogeneity to be found between schools with more or less degrees of accessibility.

Table 2: Pooled PME effect on schooling outcomes — base PSM-DiD without accessibility interaction

	2007 → 2009	2007 → 2011
<i>SAEB Mathematics</i>		
ATT	−1.891*** (0.384)	−0.882* (0.448)
R^2	0.850	0.835
<i>SAEB Portuguese</i>		
ATT	−1.399** (0.427)	−0.604 (0.447)
R^2	0.825	0.825
<i>N (Math, Port.)</i>	17,228	16,046
<i>Schools</i>	8,614	8,023
<i>IDEB (observed)</i>		
ATT	−0.137*** (0.018)	−0.068*** (0.020)
R^2	0.847	0.831
<i>N (IDEB)</i>	17,216	16,044
<i>Schools</i>	8,608	8,022

Notes: Each regression of the PSM-DiD has school fixed effects. Number of observations between IDEB and other Math/Portuguese vary slightly as some schools reported only IDEB or only reported Math/Portuguese scores. Math and Portuguese scores comprise both Ensino Fundamental (5th grade) and Ensino Médio (9th). Number of observations and school numbers are the same for Math and Portuguese scores. Standard errors are clustered at the school level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 PME & accessibility Results

As mentioned in the methods section, the heterogeneous regressions are gotten by interacting the base regression with the ISA particularly for hospitals as our main estimator, due to the widespread registration of hospitals in Brazil. The results are shown in Table 3 (with visual representation in Appendix A.20). The short term results (2007 to 2009) show similar outcomes to the base specification with one key difference. It seems that all groups have negative results in mathematics, however the *Low* accessibility groups exhibits almost double the negative coefficient as those shown by *High*, with *Mid* group in between the two in between in terms of results.

Interestingly, Portuguese results are negative and significant for Low accessibility schools, but as shown in Appendix A.22 show no statistical difference between groups. This could be explained inside the accessibility mechanism, through literature that shows that student outcomes in mathematics tend to be more responsive to teacher quality than languages (Seebruck, 2015). Therefore, until this point, the PME seems to have had two effects at work. On the one side, the PME probably had the organization/logistic problem already postulated by Almeida et al. (2016). But at the same time, more inaccessible schools also had less teacher supply, thus meaning that the same teachers would be teaching more. With decreasing marginal performance, the teachers in more isolated areas had worse performances, which had more notable effect on math scores, specially on those of more isolated schools.

In the midterm, something similar can be said. Table 3 shows that Portuguese results differences are now non-significant at the low-level, with the Appendix A.22 showing they are still not significantly different between groups. By each tier, all schools show negative coefficients, but *Low* accessibility school coefficients are non-significant. On the other hand, Math scores show a persistent negative pattern, albeit less pronounced. This could mean that the negative implementation shock had been to some extent mitigated, but the inequalities in inaccessible schools still lingered. This is seen by the still relevant difference in mathematics between *Low* and *High* accessibility groups in Appendix A.22. It is important to mention that, although the PME still had a negative impact on *High* accessibility schools, they are more than half of that experienced by the *Low* accessibility schools.

Also worth noting is that IDEB scores show a similar pattern, but it appears the lowest negative effect in the short run was in the *Mid* tier group, in the medium run still being negative but now non significant. Although results for the IDEB are descriptive, it could be showing an underlying difference between the Mid school and the others, when grades and graduation rates are combined. As a remark, it might be apparent that the

Table 3: Heterogeneous PME effects by Accessibility tier

	2007 → 2009	2007 → 2011
<i>Panel A: SAEB Mathematics</i>		
ATT (Low)	−5.636*** (0.767)	−2.539*** (0.930)
ATT (Mid)	−2.710*** (0.641)	−1.640** (0.732)
ATT (High)	−2.260*** (0.455)	−1.060** (0.526)
R^2	0.841	0.827
<i>Panel B: SAEB Portuguese</i>		
ATT (Low)	−3.269*** (0.819)	−1.342 (0.886)
ATT (Mid)	−2.100*** (0.698)	−1.680** (0.704)
ATT (High)	−2.160*** (0.503)	−1.170** (0.516)
N	15,230	14,280
Schools	7,615	7,140
R^2	0.822	0.821
<i>Panel C: IDEB (observed)</i>		
ATT (Low)	−0.194*** (0.034)	−0.113*** (0.040)
ATT (Mid)	−0.081*** (0.027)	−0.044 (0.033)
ATT (High)	−0.145*** (0.020)	−0.079*** (0.024)
N	15,220	14,280
Schools	7,610	7,140
R^2	0.843	0.825

Notes: Each row reports the absolute ATT for the corresponding accessibility tier. ATT (Low) is the coefficient on `post × treat` from the fully interacted PSM-DiD regression with school fixed effects. ATT (Mid) and ATT (High) are recovered via linear combinations of the base coefficient and the respective interaction term. Number of schools and observations are the same for Math and Portuguese scores. The ATT of this table are not a weighted ATT of Table 2 because of two reasons: 1. The sample is different as not all schools were matched to the network, leaving 683 schools out (Robustness test of this can be found in Appendix A.21) and 2. the PSM matching was done intra-tier since this was expected to lead to better controls. Standard errors are clustered at the school level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

decomposed ATTs by tiers in Table 3 do not aggregate to a weighted ATT of Table 2, this is due mainly to the choice of matching. In Table 3, the matching was done within-tier, since I theorize that the pooled matching risks connecting an isolated rural school with a highly connected urban counterpart simply because their composite probabilities are mathematically equivalent. Intra-tier matching prevents this by effectively exacting-matching on spatial structure. However, for transparency, results of a pooled matching ISA-tier interacted regression can be found in Appendix A.23, where the ATTs can be seen to be a weighted version of Table 2.

Next, I investigate whether differences in accessibility and teacher supply drive these results.

5.3 Mechanism testing

If the friction-of-implementation explanation holds, we should expect to see its signature in the schools' organizational inputs. The data permits testing one component of this, whether less accessible schools, when treated, faced a mismatch between rising student demand and the school's capacity to expand its teaching staff in response. The variables available for this test are student enrollment, the number of teachers, total staff, and the student-teacher ratio, all run through the previously specified ISA-interacted PSM-DiD model. However, I first start with testing this with pooled results, to see if it is possible from there to see any emerging trends. Appendix A.24 shows the results of these specifications. Overall, in the short and medium term, PME seems to have an increase in both Student enrollment and teacher hiring, with the student to teacher ratio increasing slightly to 0.771 students, and in the medium term to 1.01 students. Effect on staffing only seems to be relevant in the medium run. So overall, in the pooled specification there seems to be an increase in students who enroll to schools who participated in the PME, but there also seems to be an increase in teacher supply that meets that increase in demand. So from this point, there does not seem to be a mismatch. However, once the results are shown per tier, there seems to be a divergence.

Appendix A.25 shows the two different regressions accounting for heterogeneous level of accessibility to school, one categorical and one continuous. The key divergence is seen most notably in the short run. Both *Low* and *Medium* tier accessibility schools positively and significantly have a higher student enrollment, where *High* tier accessibility schools do not see this increase. On the number of teachers however, the exact opposite pattern arises. It is only the *High* accessibility tier schools who have an increase in teachers. These results hold when the regression is run with student-teacher ratios as the dependent vari-

able. Interestingly enough, staff members do not seem to rise, which could also be relevant, as the program also included that non-teachers could be part of the staff to take up the increase in new activities. So in schools where getting teachers would have been harder to get (here argued to be Low accessibility schools) also did not see any increase in them.

In the medium term, it seems that schools were finally able to adapt, and all schools increased the number of teachers they had on a similar amount. It is important to note that in the medium run, it seems medium accessibility tier schools were the ones to have the highest increase in student-teacher ratios. The continuous results of log time to nearest hospital show similar results, all else equal, a one standard deviation increase in log travel time to nearest hospital is associated with a significant 5.28 student increase, and a significant 0.197 teacher staff decrease. This being said, the increase in student-teacher ratio is only significant at the 10% level. What all this seems to suggest is that both *Low* and *Mid* accessibility tier schools had an influx of new students but had no teacher supply to meet them. This could explain in part the more negative results they had compared to the *High* tier in table 3. It remains unexplained why, even though the Mid and Low tier had similar increases in terms of student enrollment and teacher staff, the Low accessibility schools had a noticeably more negative result.

Two plausible additional channels could close the explanatory gap between similarly pressured Low and Mid tier schools. The first is teacher quality: shallower local labor markets in more isolated areas restrict the pool of candidates, so even when a school manages to expand its staff, the marginal hire is likely less qualified than in better-connected areas. The second is absenteeism: longer travel times to school correlate mechanically with higher probabilities of teacher absence, particularly in geographies where commuting itself depends on river conditions or seasonal road access. Both channels are consistent with the heterogeneity in Math results, which the literature finds to be more responsive to teacher quality than language outcomes (Seebruck, 2015). Neither is directly testable with the available data, but the directionality of both is consistent with the observed pattern: the staffing channel alone explains the existence of the gradient, while teacher quality and absenteeism plausibly explain its persistence. What the mechanism analysis establishes, then, is the gradient itself, what it suggests but cannot prove is its durability.

With the Low and Mid tier mechanisms proposed through the crowding and staffing channels, the remaining puzzle concerns the High-tier result. The mechanism operating on Low and Mid, enrollment growth without proportional teacher growth, is absent in High accessibility schools. A separate set of channels, operating through the composition of the control group and the spatial context of treated schools, is required to make sense

of the High-tier coefficient. I examine these in what follows.

5.4 Decomposing attrition

Finally, I test for differential attrition in both the base specification and across accessibility tiers. The detailed results are reported in Appendix A.26.

In the base specification, PME schools are significantly less likely to drop out of the panel than matched controls, by 5.6 percentage points at the short-term horizon (2009) and 8.9 percentage points at the medium-term horizon (2011). Because treated schools survive at higher rates, the comparison group of surviving controls is positively selected: weaker control schools attrit at higher rates, leaving a residual control sample that is stronger than the original control population. The DiD specification compares treated outcomes against this upward-biased control mean, so the resulting ATT estimates are more negative than they otherwise would be. The true magnitude of PME's negative effect on test scores is therefore likely smaller than what is reported in Sections 5.1 and 5.2. However, as the tier analysis within Appendix A.26 shows, this correction does not apply uniformly across accessibility levels.

The tier specification shows that differential attrition operates in the same direction across all tiers but is statistically significant only in the Mid and High tiers; the Low tier coefficient is small and indistinguishable from zero. This pattern carries a critical implication. Attrition bias is most severe precisely in the tier with the smallest reported negative effect, High accessibility, so correcting for it could shrink the High-tier ATT toward zero while leaving the Low-tier ATT essentially unchanged. The gap between Low and High accessibility reported in Table 3 is therefore not an artifact of differential survivorship: if anything, it is understated.

5.4.1 Mechanisms: Structural Protection and Area-Level Decline

Having established that differential attrition biases the main results anti-conservatively, the next question is *why* control schools are dropping out of the panel. Two distinct channels could account for this. The first is structural: a school's SAEB-eligible cohorts (5th and 9th grade of Ensino Fundamental) may be discontinued, either through outright institutional closure or, more commonly, through partial restructuring in which the school remains operational but no longer offers the assessed grade levels. As de Menezes et al. (2023) highlight, such consolidation decisions tend to eliminate the weakest institutions and grade offerings, mechanically inflating the surviving control group's average performance. The second is administrative: a school may remain SAEB-eligible but still

fail to validate its scores in a given year, most commonly because fewer than 80% of enrolled students attend on test day, the threshold INEP requires for results to be published (Ferrão et al., 2020). Table 4 tests both channels simultaneously, asking whether PME-treated schools behave differently from controls along each margin, and thereby isolating the channel through which the differential attrition documented above is operating (for details on the methodology of Panels A–D, see Appendix A.33).

Table 4: Attrition tests: Structural survival, administrative missingness, and spatial spillovers

	Low	Mid	High	R^2
<i>Panel A: Mechanism 1 — Structural survival in 2011</i>				
PME effect	0.063*** (0.017)	0.060*** (0.013)	0.085*** (0.010)	0.056
Pre-trend placebo (2008)	−0.003 (0.003)	0.005 (0.007)	0.014** (0.003)	0.008
<i>Panel B: Mechanism 2 — Administrative missingness Survival</i>				
PME effect	0.033 (0.028)	0.002 (0.025)	−0.011 (0.014)	0.001
<i>Panel C: Mechanism 3a — Spillover effect (Δ Math)</i>				
Hit (10 km buffer)	−3.78 (4.45)	−3.13 (4.96)	−3.26** (1.08)	0.006
<i>Panel D: Mechanism 3b — PME protective effect (Δ Math)</i>				
Treat $\times$ Hit (10 km buffer)	−3.08* (1.76)	−7.29 (8.08)	−5.67 (5.15)	0.006
matched schools (all panels)	544	1,218	6,263	
of which: Δ math observed (Panels C, D)	403	894	5,259	

Notes: Each cell reports the absolute treatment effect for the corresponding accessibility tier, recovered via the delta method from the fully interacted model. *Panel A* reports the effect of PME on the probability of structural survival (offering 5th- or 9th-grade SAEB-eligible cohorts) in 2011, with a placebo test using 2008 survival as a pre-trend check. *Panel B* reports the effect of PME on the probability of missing the SAEB Mathematics score in 2011, conditional on structural survival. *Panel C* reports the symmetric spillover effect of being within 10 km of a predicted closure on the 2007–2011 change in SAEB Math. *Panel D* reports the PME-specific moderation of this spillover effect (the protective coefficient on *treat* $\times$ *hit*). Closures are predicted from a logit model fit on the control group only, using 2007 enrollment and baseline SAEB Math scores; high-risk schools are those above the 90th percentile of predicted closure probability. All specifications use propensity-score weights and heteroskedasticity-robust standard errors. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Panel A confirms the structural survival hypothesis. PME assignment significantly reduces the probability of school closure or grade merging across all three tiers, with the largest effect in High accessibility areas (+8.5 percentage points). This gradient makes intuitive sense: in well-connected areas, students can feasibly be redirected to a nearby school, so closure is a realistic administrative option, and one that PME participation can meaningfully deter. In isolated rural areas there are far fewer alternatives nearby; closing a school would strand students with nowhere to go, making administrators reluctant to act regardless of program participation. PME’s protective role is therefore most consequential precisely where the closure option is most available. A 2008 pre-trend placebo tests whether the 2011 survival differential reflects pre-existing divergence rather than the

program. Low and Mid coefficients are near zero and insignificant. The High tier returns a small but significant 2008 estimate (+1.4 percentage points). Given that the 2007 designation of target schools in 2008 may have prompted a profilactic protection prior to funding disbursement, this divergence is more like due to early program effects than to pre-existing survival advantages. Regardless, the 2011 estimate (+8.5 percentage points) is six times larger, and the program effect remains quantitatively substantial under any reading of the placebo.

Panel B rules out the administrative missingness channel. Conditional on structural survival, treated and control schools fail to administer SAEB at statistically indistinguishable rates. The differential attrition documented in Appendix A.26 is therefore essentially structural: it flows through closure and consolidation decisions rather than through within-school test-day failures.

Panel C establishes that proximity to closure carries a cost for surviving schools. Being within 10 km of a predicted closure is associated with a decline of 3.26 SAEB Math points in the High accessibility tier ($p < 0.05$), while the effect is not significant in Mid and Low tiers, where closures are less frequent and fewer schools face this pressure. *Panel D* shows that PME does not soften this penalty: there is no statistically significant difference at the 5% level between treated and control schools in how their scores respond to nearby closures, in any tier. PME keeps schools open, but it does not insulate them from whatever is happening in their surrounding area.

The natural first reading of Panel C is mechanical reshuffling: a school closes, its students relocate to surviving schools nearby, and those receiving schools absorb a weaker intake that pulls down their average scores. Because PME disproportionately keeps treated schools open (Panel A), treated schools are the more likely recipients, 57% of treated High-tier schools lie within 10 km of a predicted closure, against 31% of controls (Appendix A.28). The mechanism is plausible, but it rests on a particular interpretation of what a closure represents, and that interpretation needs to be examined before the reshuffling story can be accepted.

Following Lehtonen (2021), school closures need not be read only as isolated events that displace students into neighbouring schools; in many contexts they are better understood as symptoms of broader population contraction, with closures and demographic decline as simultaneous expressions of the same underlying process rather than cause and effect.⁷ Under this reading, proximity to a closure is a *proxy for being located in a declining*

⁷Appendix A.29 offers correlational support for this reading: regressing schooling inputs on proximity to a closure yields a large, statistically significant decline in enrollment concentrated in the High tier. That schools near closures lose students rather than gain them is consistent with area-level contraction rather than the reshuffling of displaced students into surviving schools, though the association is descriptive and

area, and the Panel C penalty conflates two things: the demographic deterioration of the area itself, and whatever happens inside surviving schools as that deterioration unfolds. Disentangling them requires looking directly at how PME affects school inputs once the declining-area component is held fixed.

Appendix A.30 does exactly this, re-estimating the input regressions of Section 5.3 with the closure shock included as a control. The base specification had found that only Low and Mid schools gained students from PME, while High schools gained only teachers, suggesting, misleadingly, that the crowding mechanism did not operate in well-connected areas. Controlling for the closure shock overturns this. Once the declining-area component is netted out, High-tier schools also gain students from PME, and substantially so (+13.8 at 2009, +22.9 at 2011, both significant), with a correspondingly worsening student-teacher ratio (+1.1 and +1.7). The base regression had masked this because in High-tier areas two forces were pulling enrollment in opposite directions at once: PME drawing students into treated schools, and area decline draining students out of the surrounding population. The net change looked flat, which is why High-tier appeared unaffected. Separating the two reveals that PME crowds High-tier schools just as it crowds Low and Mid ones. And, it is precisely these additional students, arriving without proportional teaching capacity, that drag down the Math scores observed in Panel C. The crowding mechanism is therefore not tier-specific; it operates across the accessibility distribution, and was simply invisible in High until the area-decline confound was removed.

This reinterpretation is confirmed on the outcome side when the closure indicator is brought into the main heterogeneity specification. Appendix A.31 interacts the tier-specific ATTs from Table 3 with closure proximity, an exercise that effectively isolates schools in declining areas from those in stable ones. While short-run estimates change little, the medium-term result is decisive: the negative effect of PME on Math scores in High-accessibility schools loses significance entirely. The apparent harm to well-connected schools was driven largely by the demographic decline of their surrounding areas rather than the program itself. In schools with this much absorptive capacity, whatever genuine crowding effect remains is simply not large enough to register as a significant score loss once the local trend is removed. Crucially, this same control moves the Low-tier estimate in the opposite direction: stripping out the declining-area confound makes the Low-tier penalty larger, not smaller. Consequently, the accessibility gradient does not narrow once area decline is accounted for—it widens.

The headline negative coefficient in High-tier schools is therefore difficult to interpret as program harm. It conflates whatever genuine effect PME has on learning with the

should not be read as causal.

demographic trajectory of the areas where treated schools are concentrated, two things the unconditional estimate cannot separate. Whether the underlying effect is precisely null or modestly positive cannot be resolved with school-level data. What the evidence does seem to establish is that a meaningful share of the apparent harm to well-connected schools reflects *where those schools are located rather than what the program did to them*. The Low and Mid tiers admit no such reinterpretation: their negative effects survive the same corrections and, in the Low tier, strengthen.

These patterns carry implications for how the existing PME literature should be read. Almeida et al. (2016) acknowledge that schools dropping out of their sample tend to be lower-performing, but respond by restricting analysis to the balanced panel. This addresses the symptom without engaging with the direction of the bias: dropping weaker controls from the comparison artificially strengthens the surviving control group, and the estimated negative effect is correspondingly inflated. The tier-stratified design adopted here makes this structure visible. It additionally surfaces a spatial dimension that previous evaluations have not examined: treated schools are not randomly distributed across the territory, and the demographic trajectory of the areas they occupy shapes the counterfactual against which they are compared. Whether these biases fully account for the canonical negative PME finding, or only partially attenuate it, remains an open question, one that individual student-level mobility data would be needed to resolve. The results here are, at minimum, consistent with the interpretation that some share of the negative effects reported in the literature reflects the mechanical consequences of *where PME schools are, rather than what PME does*.

6 Robustness Check

6.1 Alternative ISA indexes

To test validity of the results of Table 3, I will run the same regressions but using two additional indexes which are similar in nature. First, I will use an index that uses both time-distance to hospitals, food centers and administrative centers (a detailed description of the weights can be found in Appendix A.9). The other specification will use the log time-cost to a hospital as a continuous variable.

The results of these tests are in Table 5, which show similar results in mathematics to the main index result: In the short term, Low accessibility schools tend to score much more negative in mathematics than Mid and High accessibility schools, and in the medium term (2011) showing that *high* accessibility schools are still more than half as negative as

Table 5: Robustness: alternative accessibility specifications

	<i>Spec. B: Weighted composite (binned)</i>		<i>Spec. C: Continuous log travel time</i>	
	2009	2011	2009	2011
<i>Panel A: SAEB Mathematics</i>				
ATT (Low) / ATT (mean)	−4.780*** (0.747)	−2.410*** (0.930)	−2.380*** (0.424)	−1.151** (0.481)
ATT (Mid)	−2.090*** (0.613)	−1.530** (0.729)		
ATT (High)	−1.520*** (0.424)	−1.000* (0.532)		
Slope (per SD log t)			−0.644*** (0.199)	−0.452** (0.229)
<i>N</i>	15,136	14,136	15,448	14,438
<i>Panel B: SAEB Portuguese</i>				
ATT (Low) / ATT (mean)	−2.930*** (0.808)	−1.160 (0.883)	−1.488*** (0.468)	−0.666 (0.500)
ATT (Mid)	−1.830*** (0.675)	−1.680** (0.696)		
ATT (High)	−1.800*** (0.479)	−1.040** (0.516)		
Slope (per SD log t)			−0.038 (0.216)	−0.214 (0.225)
<i>N</i>	15,136	14,136	15,448	14,438
<i>Panel C: IDEB (observed)</i>				
ATT (Low) / ATT (mean)	−0.186*** (0.034)	−0.099** (0.040)	−0.102*** (0.019)	−0.041* (0.022)
ATT (Mid)	−0.083*** (0.027)	−0.034 (0.033)		
ATT (High)	−0.144*** (0.020)	−0.071*** (0.024)		
Slope (per SD log t)			0.014 (0.009)	0.006 (0.010)
<i>N</i>	15,128	14,136	15,440	14,438

Notes: The composition of specification B can be found in Appendix A.9. ATTs for each tier in Specification B are reported in absolute terms, recovered via linear combinations of the base and interaction terms. Specification C matches on the overall sample and interacts the post-treatment effect with the standardized log travel time to the nearest hospital. School fixed effects, school-clustered standard errors, and propensity-score weights are applied across all models. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the *low* accessibility tier schools. Portuguese also has the same pattern as in Table 3; results are significant and more negative for Low accessibility schools in the short term. In the medium term the patterns are also the same as Table 3, the Low accessibility schools now have non significant negative results, whereas Mid and High's results are still significant.

With regards to specification C, once again a similar pattern arises, but with relevant nuances. For mathematics, in the short term, schools of all distances show a significantly negative effect of the PME program on math scores. However, by every standard deviation increase in time that it takes to get from a school to a hospital, the Math scores decrease by 0.644. In the medium term, not only more accessible schools have a lower negative impact, but each standard deviation in time diminishes to a smaller extent the negative results in Math SAEB scores. Portuguese, on the other hand, shows a similar pattern. The mean ATT is still negative and significant, but increasing the log time-distance does not have a statistically significant effect the Portuguese scores. In the medium term, effects of the PME on Portuguese scores seem to banish. This coincides, once again, with the theory of Mathematics' higher sensitivity to teacher quality than languages. Lastly, IDEB scores, in general, follow the same trends as in Table 3.

6.2 Extended sample ISA-tier specification

In addition to alternative indexes, I reran the results on the ISA tier interacted specification with the extended sample that included the ex-post connected schools mentioned in subsection 3.3. The table in the Appendix A.21 reproduces the heterogeneous specification on the extended sample, which incorporates schools that were unconnected to the multimodal transportation network in the strict sample and whose accessibility index was imputed via a nearest-neighbors procedure on geographic proximity (detailed in Appendix A.7). The exercise is a coverage robustness test: it re-incorporates precisely the kind of remote, peripherally-located schools that motivate the accessibility analysis to begin with, expanding the sample from 15,230 to 17,112 observations at 2009 and from 14,280 to 15,918 at 2011 (figure results can be seen in Appendix A.32).

The accessibility ordering survives intact. In every one of the six cells (three outcomes $\times$ two horizons), the Low-tier ATT is the most negative, the Mid-tier ATT is intermediate, and the High-tier ATT is the least negative. The spatial gradient that motivates the paper is therefore not an artifact of restricting the analysis to cleanly routable schools, it re-emerges on a sample that explicitly includes the network's periphery.

Although several coefficients are smaller in absolute value in the short term (2007 $\rightarrow$ 2009), the medium run estimates move in the opposite direction. The Low-tier coefficient on

SAEB Mathematics at 2007→2011 grows from -2.539^{***} in the main specification to -3.048^{***} here. The Mid-tier coefficient grows from -1.640^{**} to -2.641^{**} . The Portuguese Low coefficient grows from -1.342 (insignificant) to -1.890^* . The extended sample therefore tells more of a persistence story than the main analysis: when peripheral schools enter, the disruption documented at 2009 dissipates less rapidly, and the medium-run accessibility conditional effect is correspondingly larger. The main-specification estimates can in this sense be read as a conservative lower bound on the prevailing effect of PME, they understate, rather than overstate, the persistence of the implementation friction. The Mid-tier 2011 estimates for Portuguese and IDEB that lose conventional significance resemble the attenuation already visible in Table 3, and reflect the partial medium run recovery rather than a fragility specific to this robustness exercise.

6.3 Spillover and area-decline robustness

Appendix A.27 re-estimates the penalty documented in Panel C of Table 4 at 5 km and 15 km buffers, and extends the analysis to SAEB Portuguese scores. This serves two purposes: to confirm the result is robust to the choice of radius, and, more substantively, to discriminate between the two interpretations of the penalty. The penalty for the High accessibility tier is stable across radii, ranging from -3.7 to -3.9 SAEB Math points and from -3.8 to -4.6 SAEB Portuguese points, all significant at $p < 0.01$. Mid and Low tier estimates remain statistically insignificant across all radii, consistent with the argument that the channel operates primarily in denser school networks where closures and mergers are more frequent.

This stability is itself informative. A mechanical reshuffling of displaced students into surviving schools would be expected to attenuate as the buffer widens, since more distant schools are progressively less plausible recipients. The absence of any pronounced decay points to a geographically diffuse process rather than localised reshuffling: an area-level decline that affects schools throughout a contracting region regardless of their precise distance to any single closure. This does not rule out a reshuffling component entirely, as some displacement of students into surviving schools may well occur. But it appears second-order relative to the broader area effect. Thus, the buffer test does double duty. First, it confirms that the 10 km result is not an unstable boundary condition. Furthermore, its flat distance profile indicates that area-level decline, rather than reshuffling, is the principal driver of the penalty advanced in Section 5

For Portuguese scores, the pattern is different in one important way: while the initial penalty near school closures is just as large as it is for Math, the loss is partially mitigated

in PME-treated schools. This recovery does not happen for Math. The difference comes down to how school changes affect each subject. Changing schools hurts math scores much more than reading scores because math is strictly sequential; if a student misses a step in the curriculum, it is much harder to catch up (Min, 2022). Because a broken chain of math learning is harder to amend, this subject gap fits in. The broader regional decline is still real and constant across buffer choices, but its impact on actual test scores depends entirely on how fragile each subject is when local school networks shrink.

Taken together with the alternative-index and extended-sample tests, the diagnostics describe a coherent picture that qualifies the headline results differently across tiers. In Low and Mid accessibility schools the negative ATT is robust to all corrections, and the crowding mechanism documented in Section 5 provides a direct account of why PME depresses outcomes there. The High accessibility tier tells a different story. Its headline coefficient is inflated by two pressures that share a common root in area-level decline: the differential closure of weaker control schools, which positively selects the surviving comparison group, and the concentration of treated schools in declining areas, whose deterioration drags their scores independently of the program. Because proximity to a closure proxies precisely this area-level decline, conditioning on it addresses both pressures at once, and once it is accounted for the High-tier estimate becomes statistically indistinguishable from zero, particularly over the medium term. Whether that reflects a genuine null effect or a modest positive effect obscured by residual noise cannot be determined from the available data. What can be said is that the gap between Low and High accessibility, already present in the headline results, widens further once these biases are accounted for: PME's effects are starkly negative where infrastructure is thin, and consistent with approximate neutrality where it is dense.

With these tests covered, what remains is to consider what this pattern entails not only for ex-post PME evaluation, but for the design of public programs going forward.

7 Conclusion

The Amazon, geographically, in social terms, in biomes and biodiversity, is an incredibly complex and diverse region. Its transportation networks and dynamics are no different. To test the hypothesis that unaccounted-for heterogeneity in accessibility systematically underprovides for isolated schools, I constructed a comprehensive multimodal transportation network for the Legal Amazon and Brazil as a whole. This routing infrastructure operates at a massive scale, comprising 14.88 million edges and 10.93 million nodes mapped across 1.31 million kilometers of roads and over 300,000 kilometers of

navigable rivers. Although computationally expensive, this network allowed the categorization of schools into distinct accessibility tiers, and through that categorization, to ask whether the aggregate negative results documented in the PME literature were concealing heterogeneity that had gone unmeasured.

In line with previous literature, the aggregate baseline estimates demonstrate that PME had significant negative effects on SAEB mathematics scores in the short term. These effects attenuate but persist into the medium term, a pattern broadly consistent with the logistical implementation friction explained by Almeida et al. (2016). However, once the specification was interacted to account for accessibility differences, a clear divide emerged. In the short term, the mathematics penalty for isolated schools is twice as severe as for highly connected ones. While this gap appears to close in the medium term, robustness checks (Section 6) suggest this recovery is merely an illusion driven by differential survivorship, masking persistent disruptions in peripheral areas. Furthermore, this spatial inequality seems subject-specific: Portuguese scores decline uniformly across all tiers, reflecting the broader consensus that language learning tends to be more resilient to the staffing frictions that disrupt sequential math instruction.

The negative effects appear to be the product of two compounding forces. First is the logistical friction identified by Almeida et al. (2016). Second is a tier-specific amplification of that friction operating through teacher-related channels. In the short run, Low and Mid accessibility schools experienced similar increases in student enrollment with no corresponding increase in teaching staff, raising the pressure on existing teachers. The *High* tier initially appeared to behave inversely, gaining teachers but not students, an anomaly that the attrition analysis later resolves as an artifact of area-level dynamics rather than a genuine difference in how the program operated. Beyond the shared staffing pressure, I hypothesize that the larger negative impact on Math scores in the Low tier is driven by lower teacher quality, owing to shallower local labor markets, and by higher teacher absenteeism, both plausibly correlated with isolation. Testing these last two channels directly is left for future research.

This narrative is reinforced by the analysis of differential attrition. Compared to matched controls, PME-treated schools attrit at significantly lower rates. This difference is driven almost entirely by the Mid and High accessibility tiers, where school consolidation is a realistic administrative response that the program seems to deter. The surviving control sample is therefore positively selected, artificially exaggerating the negative ATTs. The key to interpreting these results lies in understanding what a school closure actually represents. Following Lehtonen (2021), closures tend to function not as isolated displacement events, but as symptoms of broader population decline; both are simultaneous expres-

sions of the same demographic process. Proximity to a closure therefore acts as a spatial proxy for this area-level contraction. By incorporating this control into the main specification to explicitly separate declining regions from stable ones, the medium-term penalty on High-tier mathematics scores loses significance entirely, while the Low-tier penalty actually grows *larger*. This control also resolves the staffing anomaly noted above: once area decline is netted out, High-tier schools are revealed to gain students from PME just as Low and Mid-tier schools do. The base regression had masked this because area decline was draining the surrounding population at the same time the program was drawing students in. The crowding mechanism is therefore not tier-specific; it operates across the accessibility distribution but was invisible in the High tier until the confound was removed. The consequence is that the High-tier negative coefficient is likely much closer to zero, or plausibly null, once the area-decline confound and differential attrition are accounted for. Though whether it is precisely zero or modestly positive cannot be resolved with school-level data. The Low accessibility tier, where neither bias is statistically detectable, emerges as the cleanest reading of PME’s effect, and the cross-tier gap documented in Section 5.2 therefore *widens* rather than narrows under correction. Insofar as the existing literature on PME, most notably Almeida et al. (2016), relies on pooled specifications that test neither for differential attrition nor for the demographic trajectory of the areas treated schools occupy, the aggregate negative effects reported there may themselves reflect, in part, a similar bias. The accessibility-stratified design adopted here makes both issues visible, and suggests that the canonical negative PME finding may mask a far more spatially heterogeneous reality than aggregate evaluations have so far recognized.

Further lines of research would extend the analysis to test heterogeneity in Blulm and Giuberti (2024)’s results for the Programa Novo Mais Educação, where the program’s emphasis fell directly on extending the day for additional math instruction. This would test whether the logistical friction explanation still holds once the program had matured, and identify the sensitivity of those results to school accessibility. A second extension would be to calculate how closely hospitals and schools are correlated with human settlements and living spaces; although granular data for smaller communities only appears officially from the 2022 IBGE census (IBGE, 2025), this could corroborate how *location-based* accessibility relates to residential proximity. Additionally, to verify the connection of school closures with area decline, the school closure regressions would be run instead with a population change estimation using gridded populations from WorldPop.

The policy implication is simple but essential: implementing a program without accounting for spatial isolation risks systematically disserve those who need it most. The diagnostics make this case even sharper. Once differential attrition and area-level decline

are accounted for, the gap between isolated and well-connected schools does not merely persist—it widens. Because the penalty in well-connected areas effectively vanishes under correction, the program’s true adverse impact on isolated schools is far more severe than the aggregate estimates suggest. The broader lesson here is that the spatial distribution of a policy’s effects matters just as much as its average. When evaluations fail to model this geography, they risk mistaking a sharply heterogeneous reality for a uniform one. Ultimately, geography and travel time are not merely variables to be controlled for; they are the current against which every policy must swim, and which may determine, long before any ministerial decree, who the program will truly reach.

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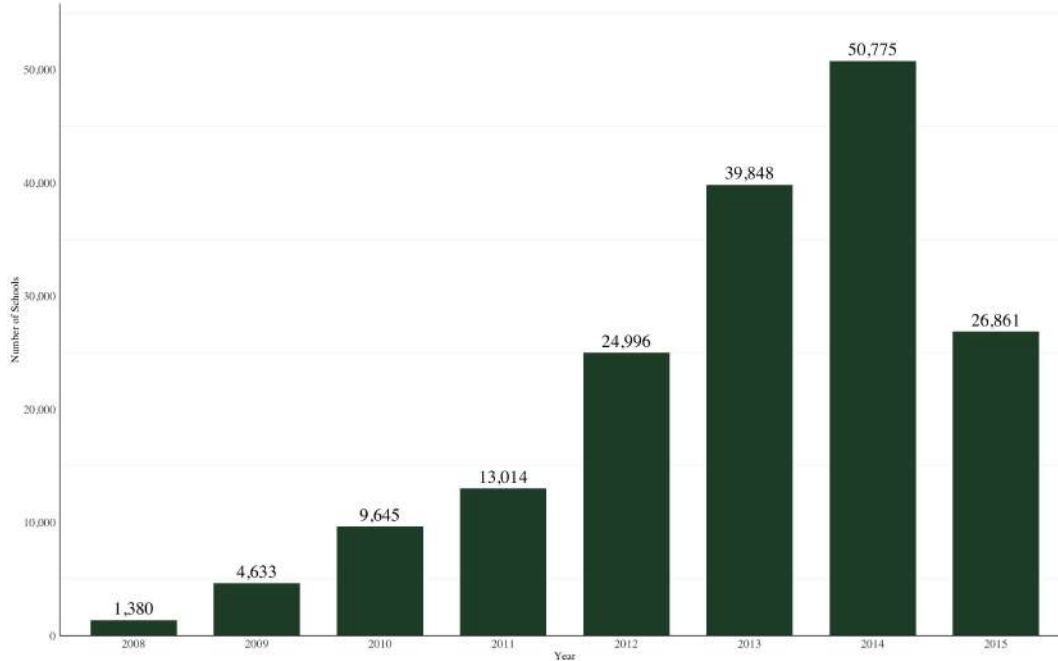
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8 Appendix

A.1 Schools receiving PME funds per year



Note: The bars represent the number of schools per year that received funding for the PME from the Programa Dinero Direito na Escola (PDDE). *Source:* Authors' elaboration based on data from the Brazilian Ministry of Education (INEP).

A.2 Fuzzy RDD design specification and results

This appendix focuses on the reproducibility exercise following the methodology done by De Oliveira and Terra (2016). This methodology tried to disentangle the effects of the PME program by taking advantage of the clearer criteria established such that schools could *then postulate to participate in the program*. The fact that the threshold was not necessarily binding is what makes the design fuzzy. The results of the first stage were found to be insignificant, therefore it seems the new criteria (mainly PBF percentage) did not seem to increase the probability of being in the PME program. Consequently, the second stage results are not considered. It is worth noting that, similarly described by De Oliveira and Terra (2016), some of the PBF percentages that were elaborated connecting the PBF database with the censo escolar's students amounts, sometimes led to percentages way above 100% (some schools had up to 3200% PBF students). It is unclear to which extent this databases could be different, and thus leading to the different results. For this specifi-

cation, the sample was restricted to schools that had a percentage of PBF students from 0 to 100%. Below are some of the specifications of the model as well as the results.

Sample. Urban schools, > 100 fundamental-education students, $\%PBF \in [0, 100]$, no PME participation in 2008–2011.

Running variable.

$$X_i = \max\{\%PBF_{i,2010}, \%PBF_{i,2011}\} - 50, \quad \text{Above}_i = \mathbf{1}_{X_i \geq 0}. \quad (6)$$

First stage. For bandwidth h :

$$\text{PME}_i = \alpha + \gamma \cdot \text{Above}_i + \beta_1 \cdot X_i + \beta_2 \cdot (\text{Above}_i \times X_i) + \varepsilon_i, \quad |X_i| \leq h. \quad (7)$$

Fuzzy IV (second stage). For outcome $Y_i \in \{\text{SAEB Math}, \text{SAEB Português}, \text{IDEB}\}$ in 2013:

$$Y_i = \tilde{\alpha} + \tau^{\text{LATE}} \cdot \widehat{\text{PME}}_i + \tilde{\beta}_1 \cdot X_i + \tilde{\beta}_2 \cdot (\text{Above}_i \times X_i) + \nu_i, \quad (8)$$

with Above_i as the excluded instrument for PME_i . Standard errors clustered at the municipality level.

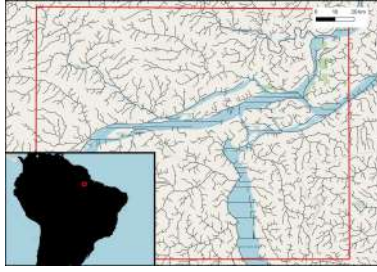
Fuzzy Regression Discontinuity at the 50% PBF priority threshold (2012 cross-section)

	Bandwidth around cutoff (percentage points)					
	$h = 10$	$h = 7.5$	$h = 5$	$h = 2.5$	$h = 1.25$	$h = 0.5$
<i>Panel A: First stage — $D = \mathbf{1}\{PME = 1 \text{ in } 2012\}$</i>						
Above $\times 1$	−0.017 (0.019)	−0.014 (0.023)	0.005 (0.026)	−0.042 (0.039)	−0.031 (0.054)	−0.176 [†] (0.095)
Above $\times X$	−0.005 [†] (0.003)	−0.001 (0.005)	0.000 (0.009)	−0.046* (0.023)	−0.030 (0.069)	−0.286 (0.297)
Observations	11,970	8,872	5,899	2,989	1,458	571
R^2	0.007	0.003	0.000	0.002	0.000	0.006
<i>Panel B: Fuzzy IV — coefficient on $\widehat{PME}$</i>						
SAEB Math	−20.70 (33.15)	−58.88 (67.12)	79.22 (2,614.7)	23.13 (77.12)	47.64 (136.2)	−162.3 (929.0)
SAEB Português	−29.38 (35.65)	−72.27 (78.66)	−1,131.1 (33,545.2)	−6.69 (58.37)	4.54 (71.84)	−236.5 (1,346.3)
IDEB	−0.275 (1.324)	−1.299 (1.981)	20.84 (662.8)	3.139 (6.133)	3.259 (8.192)	−2.195 (15.27)
<i>First-stage F</i>	1.64	1.15	0.001	0.37	0.24	0.03

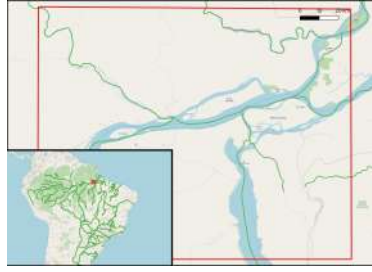
Notes: Sample restricted to urban schools with more than 100 fundamental enrolments and PBF share in $[0, 100]$. Running variable X is the centered $\max\{\%PBF_{2010}, \%PBF_{2011}\} - 50$. Standard errors clustered at municipality level in parentheses. Panel A reports the parametric first stage with linear slopes on each side of the cutoff. Panel B reports fuzzy IV second-stage estimates instrumenting PME participation with $\mathbf{1}\{X \geq 0\}$.

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

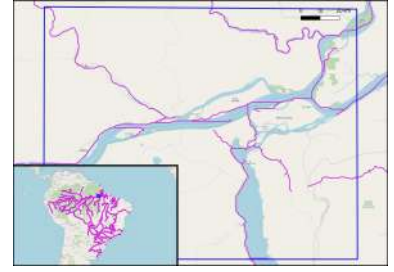
A.3 HydroSheds river processing



(a) HydroSheds raw



(b) ANTAQ database



(c) HydroSheds with ANTAQ *shadow*

A.4 Speed parameters and time penalties by edge type in the multi-modal Amazon transportation network.

Edge Type	Base (km/h)	Modifier	Speed (km/h)	Weight (min/km)
<i>Road Network</i>				
Motorway	100	+10%*	110	0.55
Trunk	80	+10%*	88	0.68
Primary	60	+10%*	66	0.91
Secondary	50	+10%*	55	1.09
Tertiary	40	+10%*	44	1.36
Residential	30	+10%*	33	1.82
Service/Living	20	+10%*	22	2.73
Track/Path	5	None	5	12.0
<i>Road Surface Modifiers</i>				
Paved	—	+10% bonus	varies	varies
Unpaved	—	−30% penalty	varies	varies
<i>River Network</i>				
Average (Moura 2017)	—	—	18.52	3.24
Maximum	—	—	37.00	1.62
<i>Transfer Connectors</i>				
POI-to-River	10	—	10	$6.0 + 10^\dagger$
Road-River crossing	—	—	—	$20^\dagger$
Informal landing	3	—	3	$2.0 + 15^\dagger$
Port/dock	5	—	5	varies

*Paved roads receive +10% speed bonus; unpaved roads receive −30% penalty or capped at 30 km/h.

[†] Fixed time penalties in minutes.

A.5 Road connector example

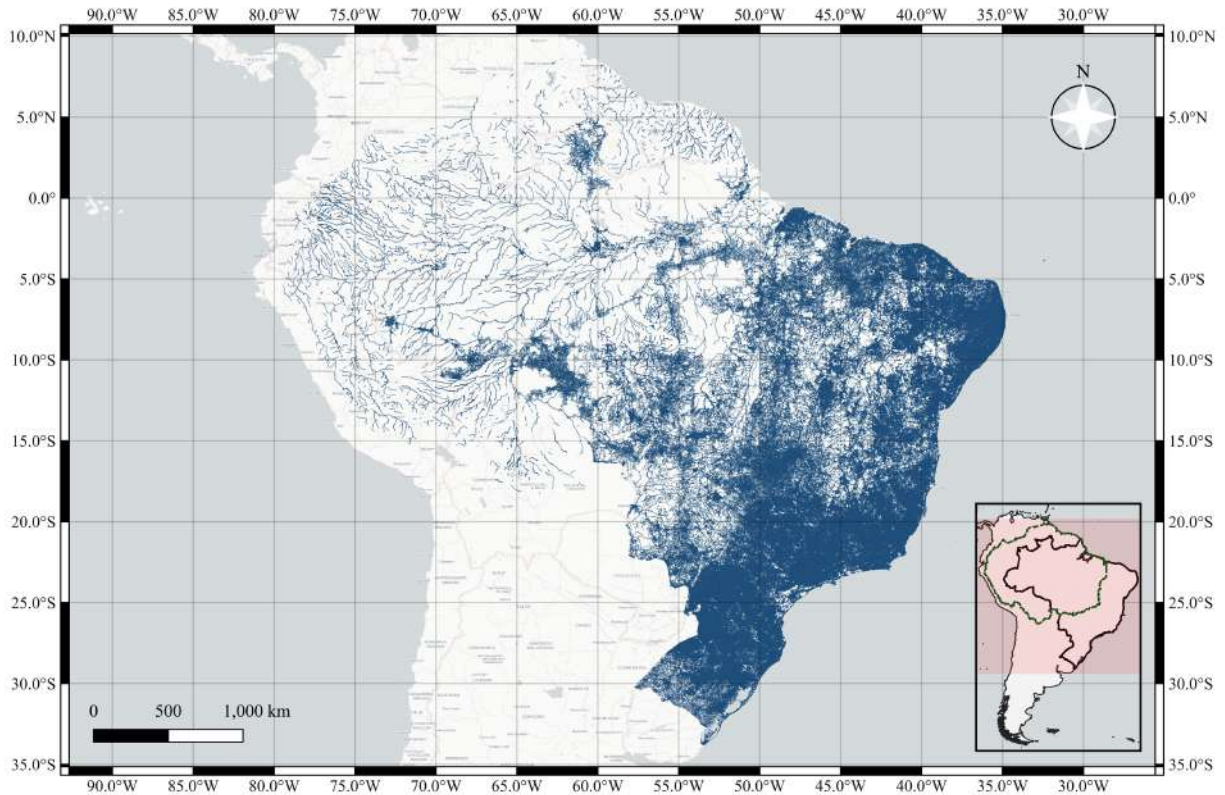


(a) Road network without bridges



(b) Road network with bridges

A.6 Brazil-Amazon graph network visualization



Note: This visualization contains all the edges used for the 3.3 section. The network for this visualization was split into a 3x3 tile GeoPackage set of files due to the computational expense of saving the whole network into a single .gpkg file.

A.7 Spatial Imputation for Off-Network Schools

A subset of schools returned $t^m = \infty$ under strict network routing due to the absence of a digitised local connector to the main road or river network—the “missing driveway” problem. A k -nearest-neighbour (k -NN) algorithm estimates off-network travel times from the local accessibility environment. Let S_i be an off-network school and $\{S_{c,n}\}_{n=1}^5$ its five nearest network-connected neighbours ordered by Euclidean distance d_n . For amenity $m \in \{\text{health, food, admin}\}$, define the local health mean:

$$\bar{t}^{\text{health}} = \frac{1}{5} \sum_{n=1}^5 t_{c,n}^{\text{health}} \quad (9)$$

Parameters:

- Neighbourhood size: $k = 5$
- Absolute isolation cutoff: $D_{\max} = 50,000$ m
- Off-trail walking speed: $v_w = 83.3$ m/min (5 km/h)
- Unmapped driveway thresholds: $d_1 \leq 3,000$ m and $\bar{t}^{\text{health}} \leq 20$ min

The spatial case is evaluated once per school using the health time environment; whichever rule is triggered applies identically to all m .

Case A — Absolute isolation. If $d_1 > D_{\max}$ or school geometry is missing, no imputation is performed:

$$t_i^m = \infty \quad (10)$$

Case B — Unmapped driveway. If $d_1 \leq 3,000$ m and $\bar{t}^{\text{health}} \leq 20$ min, the school inherits the neighbourhood mean with no walking penalty:

$$t_i^m = \bar{t}^m = \frac{1}{5} \sum_{n=1}^5 t_{c,n}^m \quad (11)$$

Case C — Off-network trekking zone. All remaining schools within $D_{\max}$ receive the nearest connected school’s travel time plus a walking-time penalty proportional to Euclidean distance:

$$t_i^m = t_{c,1}^m + \frac{d_1}{v_w} \quad (12)$$

where d_1/v_w is expressed in minutes and $t_{c,1}^m$ denotes the travel time of the absolute nearest connected school.

The number of added schools per tier of accessibility is 166, 136, 381 for Low, Mid, and High respectively. The reason High accessibility schools were added most is mostly due to them being close to an edge which is themselves not connected to a higher edge count network, and thus it remains isolated. Roads tend to be present just not in a connected manner.

A.8 Basic Health Accessibility Index

Travel Time to Nearest Health Facility (minutes)	Category	Interpretation
≤ 30	High	Good accessibility
$30 < t \leq 120$	Mid	Moderate accessibility
> 120	Low	Critical accessibility deficit

A.9 Composite Weighted Accessibility Index

Service Accessibility Scoring Rules

Service Type	Travel Time Threshold	Category	Score
Health / Food / Admin	≤ 30 min	High	2
Health / Food / Admin	$30 < t \leq 120$ min	Mid	1
Health / Food / Admin	> 120 min	Low	0

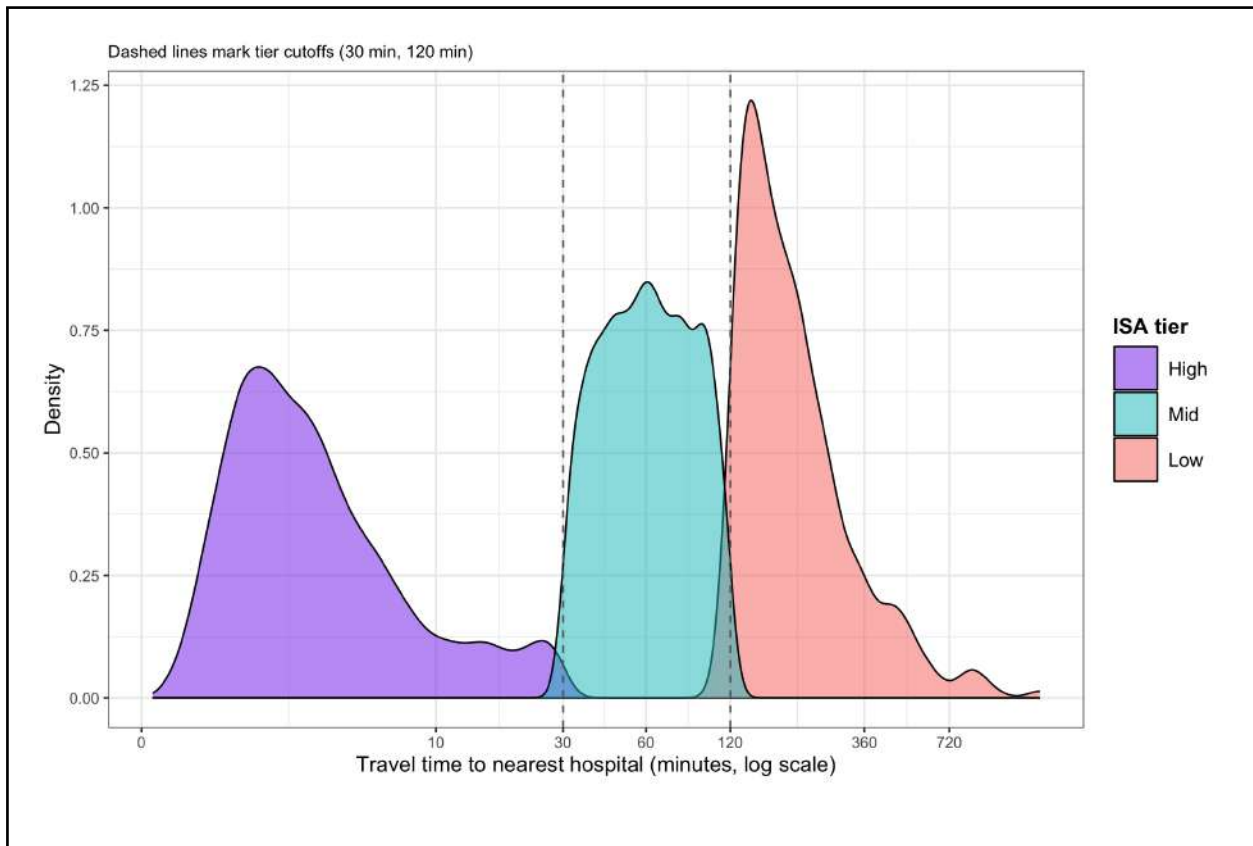
The weighted accessibility index is then calculated as:

$$\text{Access Index}_w = 0.7 \times \text{Health Score} + 0.2 \times \text{Food Score} + 0.1 \times \text{Admin Score} \quad (13)$$

where greater weight is assigned to health accessibility due to:

- the higher reliability and completeness of the health facilities database;
- the greater importance of healthcare access relative to other services.

A.10 School log of Travel time to hospital distribution across ISA tiers



Note: Apparent overlap between adjacent tier densities reflects kernel smoothing at the boundary. Tiers are defined by hard cutoffs and the underlying observations do not overlap.

A.11 Pairwise Standardized Mean Differences (SMDs) Across Tiers: Full Sample

Variable	Mid vs Low	High vs Low	High vs Mid
SAEB Mathematics (2007)	0.193	0.285	0.092
SAEB Portuguese (2007)	0.143	0.406	0.262
IDEB (2007)	0.062	0.120	0.058
Δ SAEB Mathematics (2005–2007)	0.095	−0.070	−0.165
Δ SAEB Portuguese (2005–2007)	0.082	−0.031	−0.113
Enrollment (fundamental)	−0.091	0.421	0.512
Total staff	−0.061	0.385	0.446
Classrooms (existing)	−0.006	0.139	0.145
Log municipal population	−0.066	1.160	1.220
IDHM	0.181	1.220	1.040
State school	−0.035	−0.152	−0.118
Computer laboratory	0.082	0.219	0.137
Science laboratory	0.230	0.342	0.112

Notes: Pairwise SMDs computed as the difference in tier means divided by the pooled standard deviation across all three tiers. Conventional thresholds: $|SMD| < 0.10$ negligible, 0.10–0.25 small, 0.25–0.50 moderate, > 0.50 large.

A.12 Pre-matching base specification mean testing: schools that ever participated in PME vs. schools that never participated

	Ever in PME (2008–2015)	Never in PME	Difference	SMD	p-value
<i>Outcomes & pre-trends (2007)</i>					
SAEB Mathematics	234.700	249.627	–14.926	–0.819 [‡]	< 0.001
SAEB Portuguese	223.776	235.904	–12.128	–0.734 [‡]	< 0.001
IDEB (observed)	3.316	4.010	–0.695	–0.912 [‡]	< 0.001
Δ SAEB Math, 2005–2007	–0.215	2.140	–2.355	–0.178 [·]	< 0.001
Δ SAEB Portuguese, 2005–2007	3.329	5.873	–2.545	–0.193 [·]	< 0.001
Δ IDEB, 2005–2007	0.160	0.271	–0.111	–0.189 [·]	< 0.001
<i>School size and staffing</i>					
Fundamental enrolments	583.608	513.174	70.434	0.218 [·]	< 0.001
Classrooms (existing)	12.409	13.256	–0.847	–0.057	< 0.001
Classrooms (in use)	12.394	12.872	–0.478	–0.024	0.137
Number of employees	61.023	59.907	1.115	0.030	0.017
<i>Infrastructure (share with =1)</i>					
Sports court	0.644	0.815	–0.171	–0.392 [‡]	< 0.001
Science laboratory	0.230	0.332	–0.102	–0.229 [·]	< 0.001
Computer laboratory	0.549	0.717	–0.169	–0.355 [‡]	< 0.001
Teachers’ room	0.905	0.952	–0.046	–0.180 [·]	< 0.001
DVD equipment	0.891	0.935	–0.044	–0.158 [·]	< 0.001
Public sewage	0.580	0.723	–0.142	–0.302 [‡]	< 0.001
School meals	0.938	0.950	–0.011	–0.049	< 0.001
State-administered school	0.583	0.771	–0.188	–0.410 [‡]	< 0.001
<i>Municipal characteristics</i>					
Log population (2007)	11.465	11.255	0.210	0.103 [·]	< 0.001
IDHM (2010)	0.698	0.738	–0.039	–0.558 [‡]	< 0.001
<i>N</i>	20,143	7,750			

Notes: “Ever in PME” denotes schools that participated in PME in any year between 2008 and 2015; “Never in PME” denotes schools that did not participate in any of those years. All variables measured in 2007, before any PME implementation. Difference column reports ever-PME minus never-PME. SMD denotes the standardized mean difference (Cohen’s *d*): [·] flags small imbalance ($0.1 \leq |\text{SMD}| < 0.25$), [‡] flags substantial imbalance ($|\text{SMD}| \geq 0.25$).

A.13 Fuzzy Differences-in-Discontinuity design specification and results

This appendix is dedicated to an alternative methodology that was tried previously to the FRDD methodology, and whose results work in parallel with it. Nonetheless, its results stand alone. This methodology follows Millán-Quijano (2020), and tries to answer a more specific question than the FRDD. This methodology, by combining a DiD and a FRDD looks to answer what is the effect of the formal rule (in this case the PBF 50% threshold), on the impacts of scores. So this methodology does not look to answer what the effect of the PME is on scores, but what the effect of adding this particular rule is on scores. As a consequence, identification of the effect is more restricted and smaller than what a FRDD could answer. Not surprisingly, and in a similar direction to the fact that the first stage FRDD was insignificant, the first stage of the FDiffinDisc was also insignificant. The second stage results are therefore not significant. The PBF student percentage problems are the same as those mentioned in Appendix A.2, so the sample here was also reduced to those schools who had a percentage of PBF students between 0 and 100%. A detailing of the sample, methodology and results can be found below.

Sample. Urban schools, > 100 fundamental-education students, $\%PBF_{i,t} \in [0, 100]$, observed in both pre- (2008–2011) and post-policy (2012–2015) periods.

Running variable and indicators.

$$X_{i,t} = \%PBF_{i,t} - 50, \quad \text{Above}_{i,t} = \mathbf{1}_{X_{i,t} \geq 0}, \quad \text{Post}_t = \mathbf{1}_{t \geq 2012}. \quad (14)$$

First stage. For bandwidth h :

$$\begin{aligned} \text{PME}_{i,t} = & \alpha + \gamma \cdot (\text{Above}_{i,t} \times \text{Post}_t) \\ & + \beta_1 \text{Above}_{i,t} + \beta_2 \text{Post}_t + \beta_3 X_{i,t} + \beta_4 (\text{Above}_{i,t} \times X_{i,t}) \quad |X_{i,t}| \leq h. \\ & + \beta_5 (\text{Post}_t \times X_{i,t}) + \beta_6 (\text{Above}_{i,t} \times \text{Post}_t \times X_{i,t}) + \varepsilon_{i,t}, \end{aligned} \quad (15)$$

Fuzzy IV (second stage). For outcome $Y_{i,t} \in \{\text{SAEB Math, SAEB Português, IDEB}\}$:

$$\begin{aligned} Y_{i,t} = & \tilde{\alpha} + \tau^{\text{LATE}} \cdot \widehat{\text{PME}}_{i,t} \\ & + \tilde{\beta}_1 \text{Above}_{i,t} + \tilde{\beta}_2 \text{Post}_t + \tilde{\beta}_3 X_{i,t} + \tilde{\beta}_4 (\text{Above}_{i,t} \times X_{i,t}) \\ & + \tilde{\beta}_5 (\text{Post}_t \times X_{i,t}) + \tilde{\beta}_6 (\text{Above}_{i,t} \times \text{Post}_t \times X_{i,t}) + \nu_{i,t}, \end{aligned} \quad (16)$$

with $(\text{Above}_{i,t} \times \text{Post}_t)$ as the excluded instrument for $\text{PME}_{i,t}$. Standard errors clustered at the school level.

Fuzzy Difference-in-Discontinuities exploiting the 2011→2012 introduction of the 50% PBF priority rule

	Bandwidth (pp)				
	$h = 10$	$h = 7.5$	$h = 5$	$h = 2.5$	$h = 1.25$
<i>Panel A: First stage — $D = \mathbf{1}\{\text{PME} = 1\}$</i>					
Above $\times$ 1	0.024 (0.016)	0.018 (0.018)	0.031 (0.023)	0.014 (0.035)	−0.054 (0.051)
Post $\times$ 1	0.099*** (0.020)	0.084*** (0.022)	0.077** (0.025)	0.110** (0.036)	0.065 (0.056)
Above $\times$ Post	−0.041 [†] (0.023)	−0.032 (0.026)	−0.026 (0.031)	−0.055 (0.046)	0.023 (0.071)
X slope, post	0.010*** (0.003)	0.005 (0.004)	0.001 (0.008)	0.028 (0.022)	−0.044 (0.072)
Observations	23,940	17,744	11,798	5,978	2,916
R^2	0.009	0.008	0.007	0.007	0.008
<i>Panel B: Fuzzy IV — coefficient on $\widehat{\text{PME}}$</i>					
SAEB Math	−23.09 (22.66)	−22.47 (33.16)	26.86 (47.39)	38.31 (147.2)	38.70 (121.2)
SAEB Português	−19.30 (21.28)	−22.33 (33.00)	−4.29 (36.20)	2.45 (79.52)	0.94 (61.55)
IDEB	−0.598 (0.800)	−0.599 (1.144)	1.134 (2.109)	2.049 (8.143)	−0.560 (3.007)
<i>First-stage F</i>	2.42	1.10	0.46	0.09	0.14

Notes: Stacked panel of urban schools observed in 2011 (pre-rule) and 2012 (post-rule); sample restrictions as in Appendix A.2. SAEB outcomes drawn from the 2011 wave for the pre period and 2013 wave for the post period; IDEB from the same waves. Specification follows Grembi et al. (2016): $D = \alpha + \beta \text{Above} + \gamma \text{Post} + \delta \text{Above} \times \text{Post} + f(X) + \varepsilon$, with linear slopes interacted with both Above and Post. The Above $\times$ Post coefficient (Panel A) identifies the change in the discontinuity at the 50% threshold from pre-rule to post-rule. Panel B reports the fuzzy IV second stage instrumenting PME with Above $\times$ Post following Millán-Quijano (2020). Standard errors two-way clustered at municipality and school levels in parentheses. [†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

A.14 Variables used for the PSM-DiD specification

Database	Variable	Description
<i>Outcomes (school level)</i>		
SAEB / Prova Brasil	saebMat	School-average SAEB Mathematics score
SAEB / Prova Brasil	saebPtg	School-average SAEB Portuguese score
INEP	IDEB_obs	Observed Basic Education Development Index
<i>Treatment (school level)</i>		
MEC / FNDE	pme	= 1 if school participated in PME in year t
MEC / FNDE	treat	= 1 for schools joining PME in 2008 and remaining through 2011
<i>Pre-trends (2005–2007)</i>		
SAEB / Prova Brasil	d_saebMat	Change in SAEB Mathematics, 2005–2007
SAEB / Prova Brasil	d_saebPtg	Change in SAEB Portuguese, 2005–2007
INEP	d_ideb	Change in IDEB, 2005–2007
<i>School characteristics (Censo Escolar)</i>		
INEP / Censo Escolar	QT_MAT_FUND	Number of fundamental-education enrollments
INEP / Censo Escolar	QT_SALAS_EXISTENTES	Total classrooms in the school
INEP / Censo Escolar	QT_SALAS_UTILIZADAS	Classrooms in active use
INEP / Censo Escolar	QT_FUNCIONARIOS	Total number of school employees
INEP / Censo Escolar	IN_QUADRA_ESPORTES	= 1 if school has a sports court
INEP / Censo Escolar	IN_LABORATORIO_Ciencias	= 1 if school has a science laboratory
INEP / Censo Escolar	IN_LABORATORIO_INFORMATICA	= 1 if school has a computer laboratory
INEP / Censo Escolar	IN_SALA_PROFESSOR	= 1 if school has a teachers' room
INEP / Censo Escolar	IN_EQUIP_DVD	= 1 if school has DVD equipment

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Appendix 1 – continued from previous page

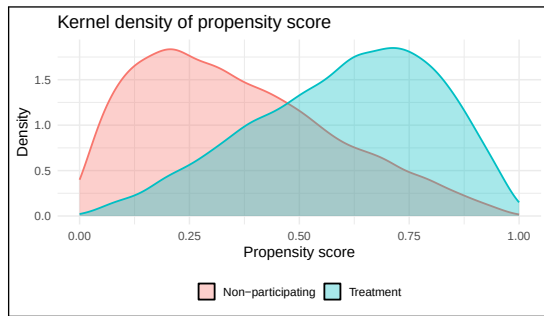
Database	Variable	Description
INEP / Censo Escolar	IN_ESGOTO_REDE_PUBLICA	= 1 if school is connected to public sewage
INEP / Censo Escolar	IN_ALIMENTACAO	= 1 if school provides meals
INEP / Censo Escolar	state_sch	= 1 if school is state-administered
<i>Municipal characteristics</i>		
IBGE	log_pop	Log of municipal population (2007 estimate)
PNUD / Atlas Brasil	IDHM	Municipal Human Development Index (2010)
<i>Accessibility (own elaboration)</i>		
Multimodal network	time_to_nearest_health_min	Travel time to nearest health facility (minutes)
Multimodal network	access_index_w	Weighted composite accessibility score
Multimodal network	index_1	Health-only accessibility tier (Low / Mid / High)

A.15 PSM Matched and Discarded observations: Base and Within-Tier Specifications

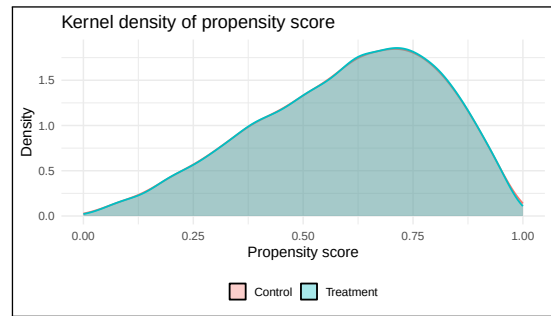
	Base PSM		Low		Mid		High	
	Control	Treated	Control	Treated	Control	Treated	Control	Treated
All	7,089	6,397	546	365	1,480	812	4,200	4,560
Matched	2,754	6,373	195	349	416	802	1,709	4,554
Unmatched	4,320	0	328	0	1,062	0	2,455	0
Discarded	15	24	23	16	2	10	36	6
% Treated retained	99.6		95.6		98.8		99.9	

Note: Nearest-neighbour propensity score matching with replacement. Base PSM pools all schools with valid covariates; within-tier PSM matches separately within each ISA accessibility tier (Low, Mid, High). “Discarded” denotes units outside the estimated propensity score common support. “Unmatched” control units lie within common support but were not selected as a match partner. % Treated retained = $\text{Matched}_{\text{Treated}} / \text{All}_{\text{Treated}}$. The treated observations between the Base and the Tier groups differ by 668 observations, this reflects mostly the network-unmatched schools.

A.16 Propensity Score Balance Comparison: before and after.



a. Before Matching



b. After Matching

A.17 Base specification Post-Matching Covariant Balance

Variable	Treated	Control	SMD	VR
<i>Pre-trends (2005–2007)</i>				
ΔSAEB Portuguese	2.611	2.315	0.022	0.975
ΔSAEB Mathematics	−0.905	−1.312	0.032	0.987
ΔIDEB	0.143	0.111	0.054	1.014
<i>Baseline outcomes (2007)</i>				
SAEB Portuguese	222.924	222.248	0.045	0.781
SAEB Mathematics	232.898	232.234	0.043	0.761
IDEB	3.178	3.149	0.041	0.634
<i>School size and staff</i>				
Enrollment (fundamental)	665.741	644.752	0.060	0.788
Classrooms (existing)	13.309	13.242	0.011	1.036
Classrooms (in use)	13.421	13.073	0.054	1.177
Staff	65.856	65.077	0.026	0.957
<i>Infrastructure</i>				
Sports court	0.669	0.657	0.025	–
Science laboratory	0.259	0.274	−0.034	–
Computer laboratory	0.541	0.561	−0.041	–
Teachers' room	0.932	0.929	0.011	–
DVD equipment	0.897	0.893	0.013	–
Public sewage connection	0.581	0.586	−0.010	–
School meals	0.940	0.940	−0.003	–
<i>School type and municipal</i>				
State school	0.534	0.548	−0.027	–
Log population	11.938	11.999	−0.032	0.613
IDHM	0.708	0.706	0.025	0.655
<i>Propensity Score</i>				
Distance (PS)	0.601	0.601	0.000	0.999

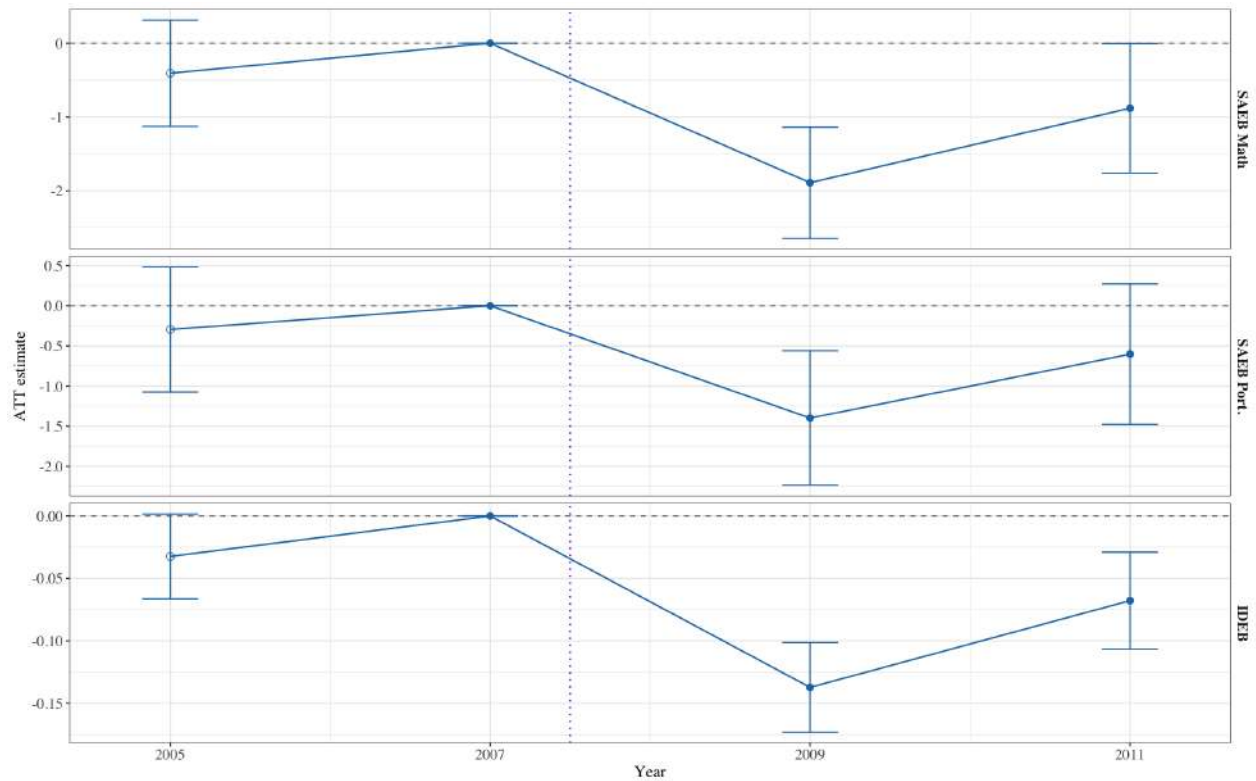
Notes: Standardized mean difference (SMD) and variance ratios (VR) reported for the matched sample using nearest-neighbor propensity score matching with replacement. Variance ratios are omitted (–) for binary indicator variables. † indicates an absolute Standardized Mean Difference ≥ 0.10 (marginal imbalance).

A.18 ISA specification Post-Matching Covariant Balance

Variable	Low Tier		Mid Tier		High Tier	
	SMD	VR	SMD	VR	SMD	VR
<i>Pre-trends (2005–2007)</i>						
ΔSAEB Portuguese	0.088	1.075	−0.051	1.046	0.077	0.948
ΔSAEB Mathematics	0.044	0.951	−0.097	0.915	0.072	0.868
ΔIDEB	−0.007	0.880	0.096	0.860	0.082	1.047
<i>Baseline outcomes (2007)</i>						
SAEB Portuguese	0.083	0.745	−0.012	0.827	0.088	0.797
SAEB Mathematics	0.030	0.758	−0.055	0.886	0.070	0.824
IDEB	0.062	0.849	0.106 [†]	0.730	0.089	0.688
<i>School size and staff</i>						
Enrollment (fundamental)	0.049	1.251	−0.072	0.610	0.092	0.757
Classrooms (existing)	0.060	1.739	−0.031	1.099	0.019	0.818
Classrooms (in use)	0.133 [†]	1.296	−0.046	1.192	0.077	0.989
Staff	0.036	1.379	−0.066	1.037	0.017	0.796
<i>Infrastructure</i>						
Sports court	0.076	–	0.028	–	−0.010	–
Science laboratory	−0.054	–	0.013	–	0.042	–
Computer laboratory	−0.041	–	0.086	–	−0.024	–
Teachers' room	−0.049	–	−0.039	–	0.014	–
DVD equipment	−0.114 [†]	–	0.027	–	−0.010	–
Public sewage connection	0.064	–	0.062	–	0.035	–
School meals	0.024	–	−0.056	–	0.022	–
<i>School type and municipal</i>						
State school	−0.040	–	−0.035	–	−0.090	–
Log population	0.091	1.330	0.069	1.011	−0.015	0.557
IDHM	−0.050	0.957	0.037	0.653	0.023	0.651
<i>Propensity Score</i>						
Distance (PS)	0.001	0.996	< 0.001	0.996	< 0.001	0.999
N Treated (Matched)	349		802		4,554	
N Control (Matched)	195		416		1,709	

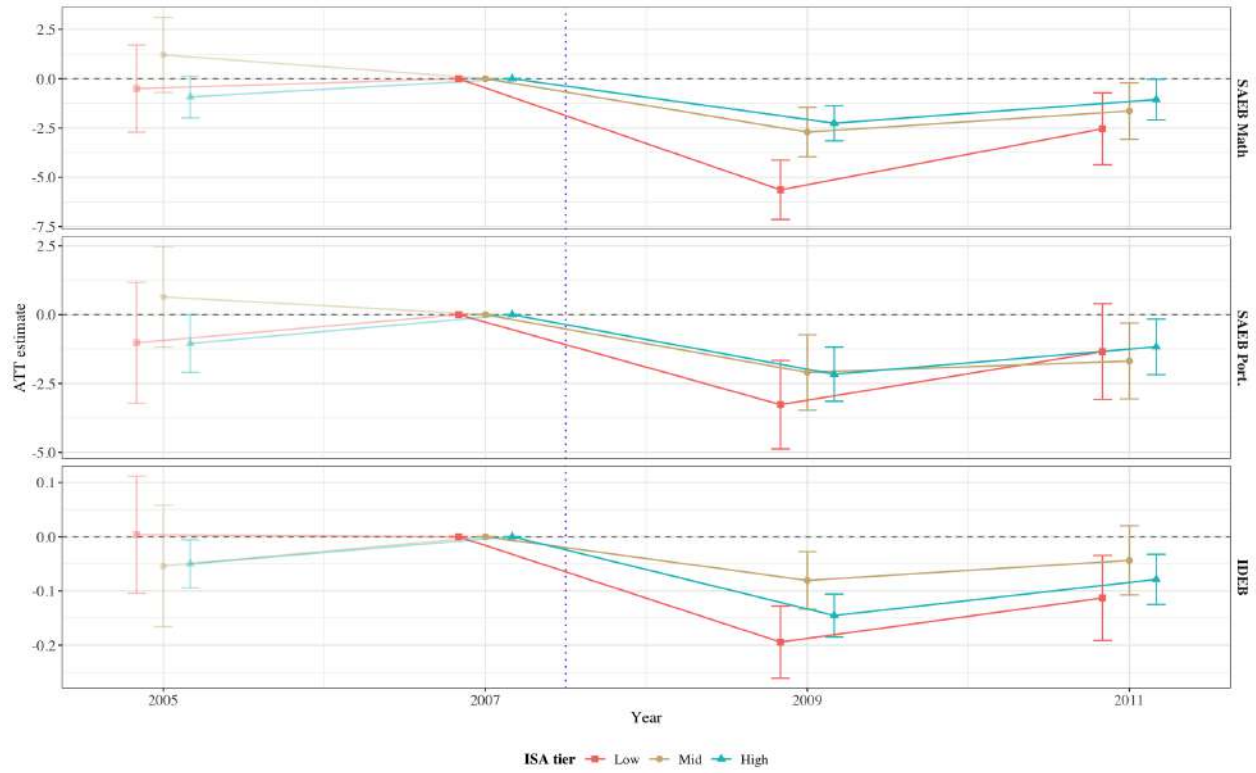
Notes: Standardized mean difference (SMD) and variance ratios (VR) reported for the matched samples across the three accessibility tiers using nearest-neighbor propensity score matching with replacement. Variance ratios are omitted (–) for binary indicator variables. N Control (Matched) reflects the number of unique control units utilized in the matched dataset. [†] indicates an absolute SMD ≥ 0.10 following Rubin (2001).

A.19 Base Specification Event Study



Note: Event study estimates from the base PSM-DiD specification (Equation 3), with 2007 as the reference period. The 2005 coefficient corresponds to the placebo test reported in Table 1 and is statistically indistinguishable from zero for SAEB Math and Portuguese, supporting the parallel trends assumption. Coefficients for 2009 and 2011 trace the dynamic ATT discussed in Section 5.1. Error bars represent 95% confidence intervals from standard errors clustered at the school level.

A.20 ISA-tier interacted specification Event Study



Note: Event study estimates from the tier-interacted PSM-DiD specification (Equation 5), with 2007 as the reference period and tier-specific ATTs recovered as linear combinations of the base coefficient and the interaction terms. The 2005 coefficients correspond to the tier-specific placebo tests reported in Table 1, which are indistinguishable from zero at the 5% level across all tiers for SAEB Math and Portuguese. The 2009 and 2011 estimates trace the dynamic ATT by accessibility tier discussed in Section 5.2. Error bars represent 95% confidence intervals from standard errors clustered at the school level.

A.21 Heterogeneous PME effects by Accessibility tier (Extended sample version)

	2007 → 2009	2007 → 2011
<i>Panel A: SAEB Mathematics</i>		
ATT (Low)	−3.539*** (0.915)	−3.048*** (1.071)
ATT (Mid)	−1.989** (0.921)	−2.641** (1.149)
ATT (High)	−1.798*** (0.491)	−1.243** (0.611)
<i>N</i>	17,112	15,918
<i>R</i> ²	0.844	0.830
<i>Panel B: SAEB Portuguese</i>		
ATT (Low)	−3.278*** (0.992)	−1.890* (1.044)
ATT (Mid)	−2.164** (1.083)	−1.692 (1.242)
ATT (High)	−1.270** (0.522)	−1.126* (0.638)
<i>N</i>	17,112	15,918
<i>R</i> ²	0.823	0.820
<i>Panel C: IDEB (observed)</i>		
ATT (Low)	−0.112** (0.046)	−0.105** (0.045)
ATT (Mid)	−0.091** (0.046)	−0.037 (0.056)
ATT (High)	−0.121*** (0.023)	−0.073** (0.030)
<i>N</i>	17,102	15,918
<i>R</i> ²	0.841	0.826

Notes: Each row reports the absolute ATT for the corresponding accessibility tier using the extended (ex-post imputed) sample. ATT (Low) is the coefficient on `post × treat` from the fully interacted PSM-DiD regression with school fixed effects. ATT (Mid) and ATT (High) are recovered via linear combinations of the base coefficient and the respective interaction term. Standard errors are clustered at the school level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.22 Heterogeneous PSM-DiD, coefficients as ISA tier differences

	2007 → 2009	2007 → 2011
<i>Panel A: SAEB Mathematics</i>		
ATT (Low tier)	−5.636*** (0.767)	−2.539*** (0.930)
Mid - Low	2.926*** (0.814)	0.900 (0.974)
High - Low	3.380*** (0.677)	1.483* (0.831)
<i>N</i>	15,230	14,280
<i>R</i> ²	0.841	0.827
<i>Panel B: SAEB Portuguese</i>		
ATT (Low tier)	−3.269*** (0.819)	−1.342 (0.886)
Mid - Low	1.171 (0.867)	−0.340 (0.923)
High - Low	1.107 (0.719)	0.172 (0.790)
<i>N</i>	15,230	14,280
<i>R</i> ²	0.822	0.821
<i>Panel C: IDEB (observed)</i>		
ATT (Low tier)	−0.194*** (0.034)	−0.113*** (0.040)
Mid - Low	0.114*** (0.035)	0.069* (0.042)
High - Low	0.049 (0.031)	0.034 (0.035)
<i>N</i>	15,220	14,280
<i>R</i> ²	0.843	0.825

Notes: Accessibility levels are interacted with the PSM-DiD estimator with *Low* being the base level. The *Mid* and *High* coefficients are thus the difference with respect to the *Low* level of accessibility. All regressions have school fixed effects. Standard errors (in parentheses) are clustered at the school level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.23 Pooled Matching Heterogeneous PME effects by Accessibility tier

	2007 → 2009	2007 → 2011
<i>Panel A: SAEB Mathematics</i>		
ATT (Low)	−5.689*** (1.256)	−0.874 (1.723)
ATT (Mid)	−1.514* (0.830)	−1.412 (1.055)
ATT (High)	−1.939*** (0.506)	−0.929* (0.553)
R^2	0.845	0.833
<i>Panel B: SAEB Portuguese</i>		
ATT (Low)	−3.626*** (1.304)	0.661 (1.683)
ATT (Mid)	−0.891 (1.013)	−1.245 (1.026)
ATT (High)	−1.377** (0.554)	−0.585 (0.591)
N	15,448	14,438
Schools	7,724	7,219
R^2	0.824	0.822
<i>Panel C: IDEB (observed)</i>		
ATT (Low)	−0.244*** (0.060)	−0.121* (0.068)
ATT (Mid)	−0.080* (0.042)	−0.031 (0.049)
ATT (High)	−0.096*** (0.022)	−0.036 (0.025)
N	15,440	14,438
Schools	7,720	7,219
R^2	0.848	0.829

Notes: Each row reports the absolute ATT for the corresponding accessibility tier. ATT (Low) is the coefficient on `post × treat` from the fully interacted PSM-DiD regression with school fixed effects. ATT (Mid) and ATT (High) are recovered via linear combinations of the base coefficient and the respective interaction term. Number of schools and observations are the same for Math and Portuguese scores. The ATTs in this table act as the direct, decomposed components of a weighted global ATT because the PSM matching was conducted on the pooled sample. However, they may differ slightly from the baseline estimates in Table 2 as not all schools were matched to the spatial network, leaving out schools without valid geographic coordinates (Robustness test of this can be found in Table A.21). Standard errors are clustered at the school level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.24 Mechanism testing: Pooled ATTs on schooling inputs

	2007→2009	2007→2011
<i>Total Enrollment</i>		
Pooled coefficient	17.800*** (4.510)	25.100*** (6.220)
R^2	0.962	0.928
N	18,148	18,090
<i>Total Teachers</i>		
Pooled coefficient	1.110*** (0.210)	1.980*** (0.286)
R^2	0.931	0.893
N	18,142	18,084
<i>Total Staff</i>		
Pooled coefficient	1.110*** 0.842* (0.432)	1.980*** 1.630*** (0.525)
R^2	0.934	0.907
N	18,148	18,090
<i>Student-Teacher Ratio</i>		
Estimate	0.771** (0.360)	1.010*** (0.382)
R^2	0.680	0.661
N	18,142	18,084

Notes: Table reports the Average Treatment Effect on the Treated (ATT) for various school-level mechanisms. Each specification includes school fixed effects. Standard errors are clustered at the school level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.25 Mechanism Analysis: Accessibility Tiers vs. Continuous Gradient

Outcome / Horizon	Accessibility Tiers (Discrete)		Log time to nearest hospital	
	2007→2009	2007→2011	2007→2009	2007→2011
<i>Total Enrollment</i>				
Low Tier	18.100** (8.270)	35.800*** (11.10)	–	–
Mid Tier	30.300*** (6.020)	52.600*** (8.510)	–	–
High Tier	8.640 (5.280)	6.450 (7.440)	–	–
Access. Gradient	–	–	5.280** (2.140)	15.400*** (2.760)
R^2	0.962	0.929	0.961	0.927
N	16,032	16,026	16,240	16,242
<i>Total Teachers</i>				
Low Tier	0.670* (0.367)	1.700*** (0.477)	–	–
Mid Tier	0.555* (0.297)	1.670*** (0.364)	–	–
High Tier	1.040*** (0.245)	1.690*** (0.304)	–	–
Access. Gradient	–	–	–0.197** (0.092)	–0.033 (0.112)
R^2	0.932	0.896	0.931	0.896
N	16,026	16,020	16,234	16,236
<i>Total Staff</i>				
Low Tier	0.565 (0.899)	0.482 (1.100)	–	–
Mid Tier	0.452 (0.742)	–0.492 (0.873)	–	–
High Tier	0.616 (0.632)	0.699 (0.724)	–	–
Access. Gradient	–	–	0.015 (0.212)	–0.264 (0.261)
R^2	0.930	0.902	0.933	0.907
N	16,032	16,026	16,240	16,242
<i>Student-Teacher Ratio</i>				
Low Tier	1.700** (0.683)	1.620*** (0.613)	–	–
Mid Tier	1.680*** (0.543)	2.070*** (0.586)	–	–
High Tier	0.781* (0.474)	0.910* (0.507)	–	–
Access. Gradient	–	–	0.235* (0.142)	0.330** (0.141)
R^2	0.666	0.644	0.685	0.657
N	16,026	16,020	16,234	16,236

Notes: Table reports ATTs across accessibility tiers and continuous gradients. Standard errors are in parentheses next to coefficients. All specifications include school fixed effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.26 Differential attrition: Rates and regression tests

	(a) Baseline Schools & Attrition Rates (%)						(b) Regression (ATT)	
	Control ($PME = 0$)			Treated ($PME = 1$)				
	Schools	2009	2011	Schools	2009	2011	2009	2011
<i>Panel A: Base specification (Pooled)</i>								
$PME_s \cdot Post_t$	2,754	8.93	16.6	6,373	4.19	10.2	−0.0557*** (0.0088)	−0.0897*** (0.0115)
R^2							0.012	0.015
<i>Panel B: Tier specification (Absolute ATTs)</i>								
ATT (Low)	195	6.67	13.8	349	4.01	10.0	−0.0258 (0.0222)	−0.0258 (0.0312)
ATT (Mid)	416	10.1	15.9	802	5.99	10.8	−0.0673** (0.0327)	−0.0511* (0.0265)
ATT (High)	1,709	7.37	13.9	4,554	3.67	9.51	−0.0450*** (0.0096)	−0.0874*** (0.0152)
R^2							0.012	0.012

Notes: Column (a) reports the raw baseline number of schools in 2007 alongside cumulative attrition rates by treatment status. Attrition is defined as a missing SAEB Mathematics score at horizon t , conditional on being observed in 2007. Column (b) reports the absolute ATT (coefficient on $PME_s \cdot Post_t$) from a weighted linear probability model on the matched sample. In Panel B, the fully interacted model uses *Low* as the reference category; standard errors for the *Mid* and *High* absolute ATTs are computed via the delta method from the joint covariance matrix. All specifications use propensity-score weights and heteroskedasticity-robust standard errors clustered at the school level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

A.27 Attrition Tests: Spatial spillover robustness

	5 km Buffer		10 km Buffer		15 km Buffer	
	Spill.	Prot.	Spill.	Prot.	Spill.	Prot.
<i>Panel A: $\Delta Math$ results</i>						
ATT (Low)	−3.08 (4.63)	−2.75 (1.77)	−3.78 (4.45)	−3.08* (1.76)	−4.35 (4.13)	−3.36* (1.78)
ATT (Mid)	−8.71 (6.10)	0.99 (8.85)	−3.13 (4.96)	−7.29 (8.08)	0.78 (3.23)	−11.1 (6.78)
ATT (High)	−3.68*** (1.22)	−2.57 (6.32)	−3.26*** (1.08)	−5.67 (5.15)	−3.94*** (1.06)	−6.90 (5.94)
<i>Panel B: $\Delta Portuguese$ results</i>						
ATT (Low)	−0.36 (3.31)	2.73 (4.02)	−0.67 (3.19)	4.17 (4.01)	−0.57 (2.99)	4.24 (3.70)
ATT (Mid)	−7.00 (5.27)	4.77 (5.55)	1.58 (4.44)	−2.71 (4.69)	2.92 (2.97)	−3.09 (3.24)
ATT (High)	−4.63*** (1.25)	4.04*** (1.33)	−4.44*** (1.08)	4.41*** (1.17)	−3.83*** (1.04)	3.68*** (1.14)
Schools			6,556			
R^2	0.006		0.006		0.007	

Notes: *Spillover* (Spill.) reports the symmetric penalty ($\widehat{Spill}_i$) of being located within the specified Euclidean radius of a predicted closure. *Protective* (Prot.) reports the PME-specific moderation of that spillover (the coefficient on $treat \times hit$). All standard errors (in parentheses) are calculated via the delta method from the joint covariance matrix. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

A.28 Compositional imbalance: exposure to predicted school closures

Tier	Group	Schools	Within 10 km of a predicted closure	
			Hit rate (%)	Mean closures
Low	Control	154	9.1	0.10
	Treated	294	13.3	0.16
Mid	Control	328	7.3	0.09
	Treated	661	20.0	0.24
High	Control	1,500	31.1	0.96
	Treated	4,285	57.0	2.40

Notes: The sample is restricted to schools with predicted closure probability below the 90th percentile of the matched sample (see Appendix A.33 for the first-stage logit specification). *Schools* reports the count of schools per group in this surviving sample. *Hit rate* is the share of schools in the group that lie within 10 km of at least one predicted closure. *Mean closures* is the average count of predicted closures within 10 km per school. Predicted closures are schools above the 90th percentile of predicted closure probability from a logit model estimated on the control group only, using 2007 enrollment and baseline SAEB Mathematics scores. All distances are Euclidean, computed in SIRGAS 2000 / Brazil Polyconic (EPSG:5880).

A.29 Attrition testing: School network shocks and labor reallocation by accessibility tier

	Low	Mid	High	R^2
<i>Panel A: Total Enrollment (Δ Enrollment)</i>				
Short term (2009)	−2.05 (18.10)	−11.70 (12.30)	−20.50*** (4.56)	0.962
Medium term (2011)	−12.10 (25.50)	−9.57 (14.40)	−53.20*** (6.22)	0.931
<i>Panel B: Teaching Staff (Δ Teachers)</i>				
Short term (2009)	1.58* (0.874)	−0.339 (0.743)	0.678*** (0.213)	0.932
Medium term (2011)	1.42 (0.956)	−1.00 (0.781)	−0.191 (0.267)	0.896
<i>Panel C: Student-Teacher Ratio (Δ Student-Teacher Ratio)</i>				
Short term (2009)	−1.24 (0.917)	0.525 (0.654)	−0.660** (0.326)	0.666
Medium term (2011)	−0.750 (1.01)	0.874 (0.784)	−0.962*** (0.358)	0.644

Notes: Each cell reports the estimated impact of proximity to a school closure (within a 10 km radius) across three distinct accessibility tiers (Low, Mid, High). Estimates are recovered via linear combinations of a fully interacted difference-in-differences specification, over short (2009) and medium term (2011) horizons. Panel A displays changes in total lower-secondary student enrollment; Panel B captures changes in the total number of contracted teachers; Panel C estimates variations in the student-teacher ratio. All specifications control for baseline school fixed effects and time trends, and are estimated using propensity-score weights. Standard errors, reported in parentheses underneath the coefficients, are clustered at the school level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

A.30 Attrition testing: Spatial Closure Effects by Tier on Schooling Inputs

Appendix 2: Mechanism testing: PME ATTs (Controlling for Closure Shocks)

	2007 → 2009	2007 → 2011
<i>Panel A: Total Enrollment</i>		
Low tier	14.400* (8.580)	28.500** (11.600)
Mid tier	27.700*** (6.450)	48.300*** (9.120)
High tier	13.800** (6.030)	22.900*** (8.420)
R^2	0.962	0.931
<i>Panel B: Total Teachers</i>		
Low tier	0.871** (0.382)	1.600*** (0.490)
Mid tier	0.725** (0.316)	1.580*** (0.382)
High tier	0.999*** (0.283)	1.660*** (0.340)
N	16,032	16,026
Schools	8,016	8,013
R^2	0.932	0.896
<i>Panel C: Student-Teacher Ratio</i>		
Low tier	1.700** (0.777)	1.800** (0.716)
Mid tier	1.720*** (0.660)	2.320*** (0.695)
High tier	1.110* (0.644)	1.650** (0.670)
N	16,026	16,020
Schools	8,013	8,010
R^2	0.666	0.644

Notes: Each cell reports the absolute average marginal effect of the PME program on the designated schooling input, isolated from the spatial drag of school closures (controlled via a closure-interaction term). Extracted via linear combinations from the fully interacted PSM-DiD model in A.25 interacted with school closures. All specifications include school fixed effects. Panel A and Panel B share the number of schools and observations. Standard errors (in parentheses) are clustered at the school level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

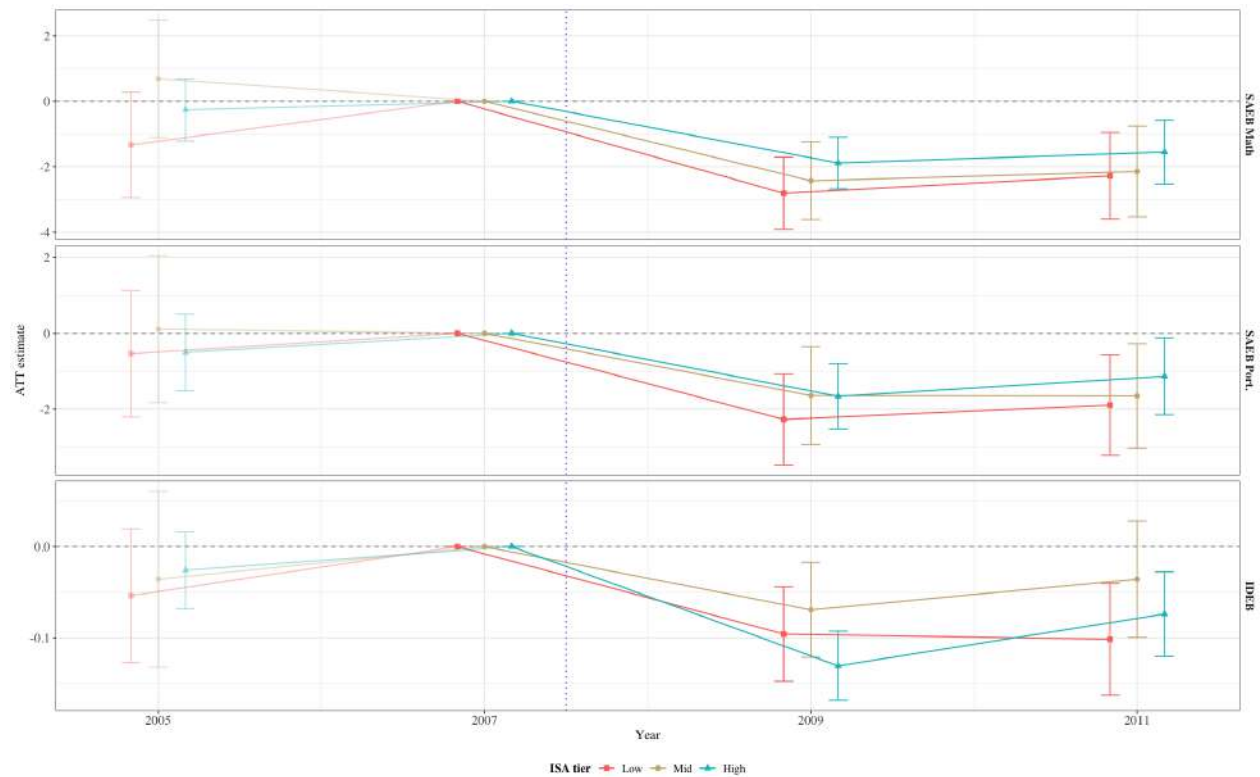
A.31 ISA tier regressions with school closure interactions

	2007 → 2009	2007 → 2011
<i>Panel A: SAEB Mathematics</i>		
ATT (Low tier)	−6.090*** (0.824)	−3.310*** (0.978)
ATT (Mid tier)	−2.810*** (0.739)	−1.710** (0.821)
ATT (High tier)	−2.240*** (0.588)	−0.967 (0.644)
<i>N</i>	15,230	14,280
Schools	7,615	7,140
<i>R</i> ²	0.842	0.828
<i>Panel B: SAEB Portuguese</i>		
ATT (Low tier)	−4.080*** (0.882)	−2.560*** (0.953)
ATT (Mid tier)	−2.560*** (0.780)	−2.150*** (0.784)
ATT (High tier)	−2.470*** (0.624)	−1.520** (0.630)
<i>N</i>	15,230	14,280
Schools	7,615	7,140
<i>R</i> ²	0.822	0.822
<i>Panel C: IDEB (observed)</i>		
ATT (Low tier)	−0.239*** (0.037)	−0.168*** (0.042)
ATT (Mid tier)	−0.111*** (0.031)	−0.063* (0.035)
ATT (High tier)	−0.135*** (0.025)	−0.100*** (0.027)
<i>N</i>	15,220	14,280
Schools	7,610	7,140
<i>R</i> ²	0.844	0.825

Notes: Each cell reports the absolute Average Treatment Effect on the Treated (ATT) for the corresponding accessibility tier, extracted via linear combinations. All specifications include school fixed effects. Standard errors (in parentheses) are clustered at the school level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.32 ISA-tier interacted with extended sampe specification Event Study



Note: Event study replicating the tier-interacted PSM-DiD specification on an extended sample that includes schools that were not directly matched directly in the network, and whose accessibility indexes were calculated *ex-post*. The 2007 reference period and recovery of tier-specific ATTs follow the same procedure as in Appendix A.20. The qualitative pattern of larger negative effects in Low accessibility schools holds in the extended sample, with attenuated magnitudes overall, consistent with the inclusion of schools whose treatment status or outcome variability differs from the matched sample used in the main specification. Error bars represent 95% confidence intervals from standard errors clustered at the school level.

A.33 Attrition Tests: Specifications and Estimation

This appendix details the specifications behind Table 4 (Panels A–D).

A.33.1 Sample and notation

I run all mechanism tests on our final propensity-score-matched sample ($N = 8,025$ schools), using the tier-specific weights ω_i generated in Section 4.2. Standard errors are heteroskedasticity-robust. Notation:

- $D_i \in \{0, 1\}$: PME treatment status (fixed at 2007 baseline).
- $T_i \in \{\text{Low, Mid, High}\}$: accessibility tier; Low is the omitted category.
- ω_i : tier-specific PSM weight.
- $\Lambda(\cdot)$: logistic CDF.

A.33.2 Mechanism 1 (Panel A): Did PME prevent structural closures?

Outcome. We define a school as "surviving" ($S_{i,y}$) if it appears in the Censo Escolar offering its standard grade levels:

$$S_{i,y} = \mathbb{1}\{\text{IN_FUND_AI}_{i,y} = 1 \vee \text{IN_FUND_AF}_{i,y} = 1\} \quad (17)$$

where IN_FUND_AI and IN_FUND_AF indicate offering *anos iniciais* (5th grade) and *anos finais* (9th grade), respectively. Schools missing from the year- y Censo are coded as 0.

Specification. We estimate the probability of survival in 2011 using a weighted linear probability model. We fully interact the PME treatment (D_i) with the accessibility tiers (T_i) to measure how survival rates vary by geography:

$$S_{i,2011} = \alpha + \beta D_i + \sum_t \gamma_t \mathbb{1}T_i = t + \sum_t \delta_t (D_i \times \mathbb{1}T_i = t) + \varepsilon_i \quad (18)$$

The absolute treatment effect for each tier is simply the base effect plus the tier-specific interaction:

$$\hat{\tau}_{\text{Low}}^{\text{M1}} = \hat{\beta}, \quad \hat{\tau}_{\text{Mid}}^{\text{M1}} = \hat{\beta} + \hat{\delta}_{\text{Mid}}, \quad \hat{\tau}_{\text{High}}^{\text{M1}} = \hat{\beta} + \hat{\delta}_{\text{High}} \quad (19)$$

Pre-trend placebo. To ensure we are not just capturing schools already on a positive trajectory, we re-run this model for 2008 (before PME was fully rolled out). These placebo

effects ($\hat{\tau}_t^{\text{M1,placebo}}$) are practically zero, with a minor exception in the High tier noted in Section A.33.2.

A.33.3 Mechanism 2 (Panel B): Did open schools dodge the exam?

If treated schools stayed open, did they actually administer the SAEB exam, or did they strategically avoid it?

Outcome. We restrict the sample to schools that survived until 2011 ($S_{i,2011} = 1$) and flag those missing a math score:

$$M_i = \mathbb{1}_{\text{saebMat}_{i,2011} \text{ is missing}} \quad (20)$$

Specification. We run the same interacted model, but strictly on the surviving subpopulation:

$$M_i = \alpha' + \beta' D_i + \sum_t \gamma'_t \mathbb{1}_{\{T_i=t\}} + \sum_t \delta'_t (D_i \times \mathbb{1}_{\{T_i=t\}}) + \nu_i \quad (21)$$

Interpretation. Because survival is directly influenced by treatment (as proven in Mechanism 1), restricting the sample breaks random assignment. Therefore, this test is purely descriptive. Conditional on survival, we find no differential missingness, though selection into survival prevents a strict causal interpretation

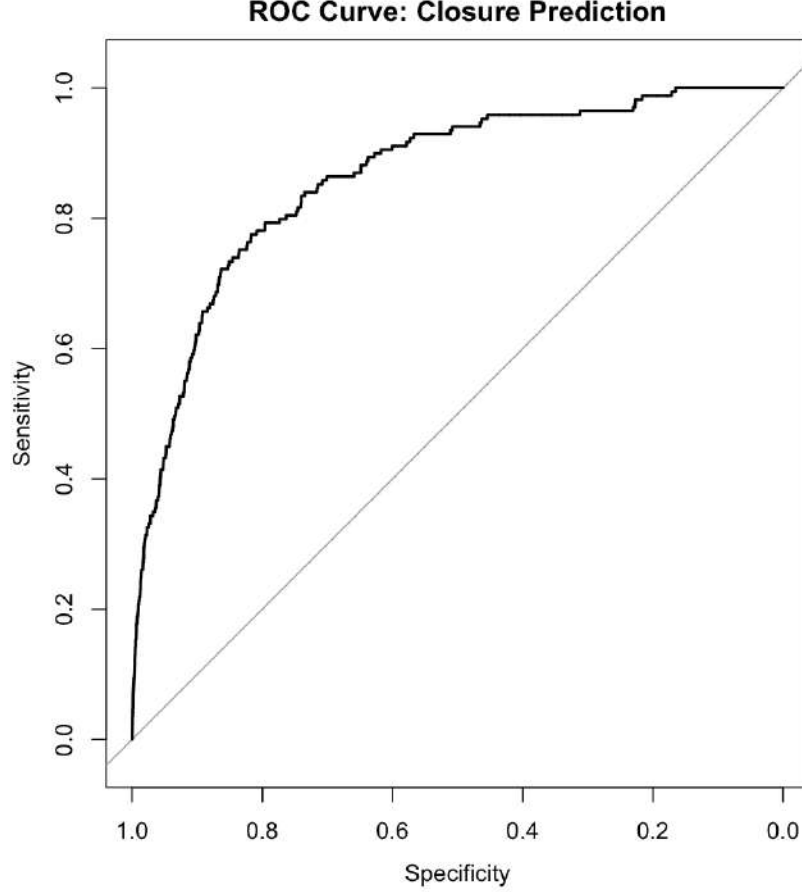
A.33.4 Mechanism 3: Spatial spillovers

When a school closes, displaced students crowd into nearby open schools. To measure this "spillover" penalty without introducing bias, we need three steps: predicting closure risk, defining the spatial "blast radius," and running an interacted regression.

Step 1: Predicting closures Using actual closures would bias our results because PME directly prevented closures. Instead, we train a logistic regression exclusively on the control group ($\mathcal{N}_0$) to predict closure risk based strictly on 2007 baseline data:

$$\Pr(S_{i,2011} = 0 \mid X_i,) = \Lambda(\pi_0 + \pi_1 \text{enrol}_{i,2007} + \pi_2 \text{saebMat}_{i,2007}) \quad (22)$$

Predictive Performance of Baseline Institutional Distress Model



Note: Receiver Operating Characteristic (ROC) curve evaluating the predictive performance of the first-stage logistic regression (Equation 22). The model utilizes 2007 baseline enrollment and SAEB mathematics scores to classify schools into closure risk percentiles. The steep initial trajectory indicates a high true positive rate (Sensitivity) relative to the false positive rate (1 - Specificity), yielding an Area Under the Curve (AUC) that supports the model's validity as a robust detector of severe institutional distress prior to PME implementation.

Risk classification. Schools in the top 10% of predicted risk ($\hat{p}_{(0.90)}$) are classified as "predicted closures" ($\mathcal{C}$). The remaining 90% serve as our spatial analysis sample ($\mathcal{S}_{\text{exo}}$):

$$\mathcal{C} = i : \hat{p}_i \geq \hat{p}_{(0.90)}, \quad \mathcal{S}_{\text{exo}} = i : \hat{p}_i < \hat{p}_{(0.90)} \quad (23)$$

Step 2: Defining spatial exposure We draw a 10 km Euclidean radius around every school in our analysis sample. A school is considered "Hit" if it falls within 10 km of a predicted closure:

$$\text{Hit}_i = \mathbb{1} \{ \exists j \in \mathcal{C} : \|\text{loc}_i - \text{loc}_j\|_2 \leq 10 \text{ km} \} \quad (24)$$

(Robustness checks for 5/15/20,km radii and continuous closure counts are in Appendix A.27).

Step 3: Outcome regression (Panels C and D) Outcome. We use the change in math scores from 2007 to 2011. Differencing removes static geographic disadvantages:

$$\Delta Y_i = \text{saebMat}_{i,2011} - \text{saebMat}_{i,2007} \quad (25)$$

Specification. We run a fully saturated model interacting Treatment (D_i), Spillover Exposure (Hit_i), and Tier (T_i):

$$\begin{aligned} \Delta Y_i = & \alpha'' + \beta'' D_i + \theta \text{Hit}_i + \sum_t \gamma_t'' \mathbb{1}_{\{T_i=t\}} \\ & + \sum_t \delta_t'' (D_i \times \mathbb{1}_{\{T_i=t\}}) + \sum_t \phi_t (\text{Hit}_i \times \mathbb{1}_{\{T_i=t\}}) \\ & + \rho (D_i \times \text{Hit}_i) + \sum_t \psi_t (D_i \times \text{Hit}_i \times \mathbb{1}_{\{T_i=t\}}) + \nu_i \end{aligned} \quad (26)$$

Panel C: The symmetric spillover penalty. First, we calculate the general penalty of being near a closure, regardless of PME status. This captures the compositional drag of absorbing displaced students:

$$\widehat{\text{Spill}}_{\text{Low}} = \hat{\theta}, \quad \widehat{\text{Spill}}_{\text{Mid}} = \hat{\theta} + \hat{\phi}_{\text{Mid}}, \quad \widehat{\text{Spill}}_{\text{High}} = \hat{\theta} + \hat{\phi}_{\text{High}} \quad (27)$$

Panel D: The PME protective effect. Next, we isolate the differential effect of PME to see if treated schools mitigated the spillover shock better than control schools in the exact same blast radius:

$$\widehat{\text{Protect}}_{\text{Low}} = \hat{\rho}, \quad \widehat{\text{Protect}}_{\text{Mid}} = \hat{\rho} + \hat{\psi}_{\text{Mid}}, \quad \widehat{\text{Protect}}_{\text{High}} = \hat{\rho} + \hat{\psi} * \text{High} \quad (28)$$

A.33.5 Connecting the math to the intuition

The complex regression in Equation (26) boils down to a simple difference-in-differences logic. For any tier t , the "Protective Effect" is simply the performance gap between treated and control schools inside the blast radius, minus the gap between treated and control schools in safe zones:

$$\widehat{\text{Protect}}_t = \widehat{\text{ATT}}_{t,\text{hit}} - \widehat{\text{ATT}}_{t,\text{unhit}} \quad (29)$$

A.33.6 Sample sizes across the mechanism pipeline

Our sample naturally shrinks as we apply stricter analytical conditions:

- **Panels A–B (Matched sample):** $N = 8,025$. All tier-matched schools.

- **Panel B (Mech 2 subsample):** $N = 7,844$. Conditional on surviving to 2011; the 181-school drop equals the number of structural closures detected by Equation (17).
- **Panels C–D (Mech 3 subsample):** $N = 6,556$. Conditional on surviving the 90th-percentile risk threshold and possessing endpoint scores. This 1,469-school drop corresponds exactly to the attriters documented in Table 4.

A.33.7 Standard error treatment

All regressions use heteroskedasticity-robust standard errors. When we calculate tier-specific ATTs by adding coefficients together, we use the delta method to compute the correct joint standard errors.⁸

⁸For this I use `marginaleffects::hypotheses()` in R.